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Wave-Corrected Caustics over Photon Splatting: Fold Kinematics, a Band-Local Airy Correction, and a Comparative Evaluation Against Production Solvers

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Abstract

Photon-splatting and path-traced caustics are the standard way to render the light focused by a curved transparent body onto a diffuse receiver. Both families evaluate a geometric push-forward whose Jacobian determinant vanishes on the fold locus, so the estimator diverges there, and no amount of sampling repairs it. We state that failure as a property of the model rather than of the discretisation, then describe a hybrid estimator in which a geometric quadrature carries the whole image and a wave-optical correction is applied only inside a narrowly detected fold band: the band's own geometric energy is redistributed according to the Airy fold profile $|\mathrm{Ai}(-s/w)|^2$*, with a per-channel width* w_c *following the* $\lambda^{2/3}$ *scaling implied by the fold's canonical form. We first derive the kinematics from scratch — the ray map, the fold condition, the normal form of the fold and the wavelength scaling that follows from it — and then use that derivation to explain where the standard computer graphics constructions for caustics succeed and where they necessarily fail.*

We evaluate against PBRT-v4 SPPM, Blender Cycles and Mitsuba 3 on an identical scene with measured dispersive indices (water: Daimon & Masumura 2007; glass: Rubin 1985). Against an independent brute-force geometric ground truth of 810,000 *explicit rays, our estimator scores* 0.917 *correlation at 40 mm tiles against* 0.853 *(SPPM) and* 0.828 *(Cycles); at 5 mm tiles the ordering is* 0.721 / 0.722 / 0.405*. The converged two-dimensional shape of our field agrees with SPPM at* $r = +0.932$*, and its transport throughput measures* 96.41% *at three wavelengths with Fresnel reflection as the only loss channel.*

We are explicit about what the method does not *establish. The correction is defined only for a fold; it says nothing about higher-order caustics, and we show by direct measurement that a naive normalisation of the band is* impossible *—* $|\mathrm{Ai}(-u)|^2$ *is not integrable on* $\mathbb{R}$ *— which is why the correction is written as a discrete band ratio rather than a continuous constant. Two of our own earlier claims are withdrawn in this paper, each killed by an estimator artefact rather than by the physics, and both are documented because they are the failure mode we expect a reader to hit.*

Keywords: caustics, wave optics, photon splatting, photon mapping, SPPM, diffraction, Airy function, grid-free rendering, comparative evaluation

1. Introduction

A caustic is the envelope of a family of refracted or reflected rays. Where the envelope touches the receiving surface, neighbouring rays cross, the surface-to-receiver map loses rank, and the ray density that a renderer integrates becomes infinite. Every production caustic method in use today integrates that density. The methods differ in how they estimate the integral — photon maps store the ray density as points [Jen96], SPPM does so with a radius that shrinks as it converges [HOJ08], path tracers connect vertices [Vea97] — but they share the integrand, and the integrand is singular on the very feature the user is trying to see.

This paper starts from that observation. The first direction is theoretical, and it is the reason the paper opens with kinematics rather than with rendering. The singularity is not an accident of any parameterisation. It is the fold of a smooth map, it has a normal form, and the normal form dictates both how the geometric estimate fails and what the correct description near the fold looks like. It also fixes a number: the fold is described near its degeneracy by a cubic phase, and the cubic phase carries one wavelength and one geometric coefficient, whose ratio is a length. That length scales as $\lambda^{2/3}$, not as λ and not as the Fresnel $\sqrt{\lambda}$. Nobody fits that exponent; it falls out of the normal form, and it is what makes a per-channel fringe width more than a decoration.

The second direction is practical. If the wave description is right near the fold, and the geometric description is right away from it,

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then a single estimator can use each where it is valid, at the cost of detecting where the boundary is. We implement that estimator and then spend the second half of the paper trying to falsify it against the tools that practitioners actually run. The evaluation is deliberately unflattering in its construction: the ground truth is an independent brute-force trace that shares no code with the method under test, the reference conforming solver is PBRT-v4 SPPM [PJH23], and a path-traced comparison is produced by Blender Cycles on the same scene with the same measured dispersive indices. Where our method loses, we say so.

1.1. Why the kinematics come first

It is possible to write a caustic renderer without ever writing down the map that is being inverted, and most of the literature discourages doing so: photon mapping is presented as a Monte Carlo estimator over paths, and its convergence is discussed in terms of sample counts and radii. That presentation is efficient, but it hides the one property that decides whether the method can work at all. A density estimator whose target integrand diverges is not a convergence-rate problem. Increasing the sample count does not help, reducing the kernel radius does not help, and adding a denoiser hides the failure behind a smooth blur that no longer means anything. The only useful question is whether the estimator's support contains the singular set, and that question is answered by the rank of a Jacobian.

So we write the kinematics down. Section 2 defines the surface, the ray map, and the exact irradiance it induces, which is the push-forward of the surface measure. The fold locus is then the zero set of the Jacobian determinant, and we prove the two statements the rest of the paper leans on: that a geometric quadrature cannot converge near the fold at any resolution (Lemma 3), and that the correct local description is the canonical fold integral whose amplitude envelope is the Airy function (Proposition 5). From the same normal form we extract the wavelength scaling (Corollary 6) and, because we have been bitten by this, the statement that the resulting intensity is not integrable - so a correction built on it cannot be normalised by a constant (Lemma 7).

1.2. How computer graphics builds caustics

Section 3 then reads the standard constructions as answers to the kinematic problem, which makes their shared limitation visible. Forward constructions — photon mapping and its progressive variant — estimate the push-forward by depositing samples and integrating density. Backward constructions — path tracing, and the bidirectional and vertex-connection variants [Vea97, MGN17] — estimate it by connecting vertices and dividing by a geometric term. Both factor the same $1/|\det J_{\mathbf{H}}|$, and both become invalid exactly on $\mathcal{F}$. The section states where each family is exact, where each is merely expensive, and where neither can be repaired, because that boundary is what the hybrid estimator is built around.

1.3. The current way, and the way we take

Sections 4 and 5 are the two halves of the comparison. The first measures what current practice produces and costs. We take the position that a comparative claim is only meaningful if the comparison runs on one scene, with one set of measured material data, one source, and one set of estimators applied identically to all sides; the section therefore reports both the numbers and the estimator convention that produced them, including one previously published statistic that we withdraw because it was an artefact of that convention (Section 7.7).

The second half states our method, which is small: keep the geometric quadrature, detect the fold band analytically from $|\det J_{\mathbf{H}}|$, and inside the band replace the divergent density with a normalised Airy profile while conserving the band's own measured flux. The method introduces three quantities — a detector percentile, a reference fringe width, and a ceiling — and Section 6.7 measures the sensitivity to each rather than asserting a value.

1.4. Contributions

- A kinematic account of the caustic fold that connects the rendering estimator to the fold's normal form, with the $\lambda^{2/3}$ fringe scaling derived rather than fitted (Section 2).
- The observation, proved and measured, that the Airy fold intensity is not integrable, so a flux-conserving correction must be formulated as a discrete band ratio; a normalisation constant does not exist (Lemma 7, Section 5.3).
- A hybrid estimator, its implementation, and an ablation over its three parameters (Sections 5, 6.7).
- A like-for-like evaluation against PBRT-v4 SPPM, Blender Cycles and Mitsuba 3 on measured dispersive materials, with the estimator conventions stated and the losses reported (Sections 6, 7).
- Two withdrawn claims from our own earlier drafts, each traced to a measurable estimator artefact, documented so the failure mode is reproducible (Sections 7.6, 7.7).

1.5. What this paper does not claim

The correction applies to folds. It has no content for higher-order caustics — cusp points, swallowtails, or the Pearcey and umbilic normal forms that govern them — and we do not have a tested implementation for those cases (Section 8). We also do not validate against a physical apparatus. The comparison in this paper is entirely computational, against solvers and against an independent trace, and the quantities compared are therefore properties of those solvers and that scene. A laboratory measurement would test a different claim, and the scene here is specified precisely enough that someone may make it.

1.6. Reading order

Section 2 is self-contained and can be read as a short treatment of fold optics for graphics. Readers who want only the method should read Section 5 after Section 3; readers who question the numbers should read Section 2.12 first, because two of our earlier errors lived there and not in the physics.

2. Kinematics of the fold

This section is the foundation the rest of the paper stands on. It defines the scene, derives the map from surface positions to receiver positions, states exactly what irradiance that map induces, and then establishes the statements about the fold that matter for rendering: that its density is unbounded, that it has a normal form, that the normal form yields the wavelength scaling, and that the resulting intensity cannot be normalised by a constant.

2.1. The scene

The test configuration is a circular glass cylinder, a water column inside it, and a matte receiver, lit by a distant source. The parameters are fixed for the whole paper and are listed in Table 1; Figure 1 draws the arrangement to scale.

2.2. The surface and its normal

Let the water surface be a height field over $\mathbf{x} = (x, y) \in \mathbb{R}^2$,

$$h(\mathbf{x}, t) \;=\; \sum_{i=1}^{4} A_i \sin\big(k_i\,(\hat{\mathbf{u}}_i \cdot \mathbf{x}) - \omega_i t\big), \qquad k_i = \frac{2\pi}{\Lambda_i}, \tag{1}$$

a sum of four directional sinusoids with amplitudes $A_i = \{0.65, 0.40, 0.25, 0.15\}$ mm at wavelengths $\Lambda_i = \{42, 27, 17, 11\}$ mm. The expansion in (1) is convenient rather than essential: nothing below uses the sinusoidal form except to know that ∇h is analytic and that $|\nabla h|$ is small. The outward normal follows from the analytic gradient,

$$\mathbf{n}(\mathbf{x}, t) \;=\; \frac{(-\partial_x h,\; -\partial_y h,\; 1)}{\sqrt{1 + |\nabla h|^2}}. \tag{2}$$

For the amplitudes above, $|\nabla h| \le \sum_i A_i k_i \approx 0.34$ at the shortest wave, so the surface stays well inside the small-slope regime in which the paraxial normal form of Section 2.6 is the leading term. This matters because the fold normal form is asymptotic, and a surface steep enough to produce higher-order caustics would leave its domain of validity. We return to this in Section 8.1.

2.3. The ray map

Let the source direction be the unit vector $\hat{\mathbf{s}}$ pointing along propagation (downward, into the surface). Vector Snell refraction at the interface gives the transmitted direction. With the ratio $\eta = n_1/n_2$ and the cosine of incidence $c_i = -\hat{\mathbf{s}} \cdot \mathbf{n} > 0$,

$$\hat{\mathbf{t}} \;=\; \eta\,\hat{\mathbf{s}} + \big(\eta\, c_i - \sqrt{1 - \eta^2(1 - c_i^2)}\big)\,\mathbf{n}, \qquad \eta^2(1 - c_i^2) \le 1, \tag{3}$$

where the inequality failing signals total internal reflection and the sample carries no transmitted power. A refracted ray leaves the surface at height $h(\mathbf{x})$ and reaches the receiver plane at $z = -D$ after a path length $s = (-D - h(\mathbf{x}))/\hat{t}_z$. Composing the two gives the *surface-to-receiver ray map*

$$\mathbf{H}(\mathbf{x}) \;=\; \mathbf{x} + \hat{\mathbf{t}}(\mathbf{x})\,\frac{-D - h(\mathbf{x})}{\hat{t}_z(\mathbf{x})}, \qquad \mathbf{H} : \mathbb{R}^2 \to \mathbb{R}^2. \tag{4}$$

Write $J_{\mathbf{H}} = \partial \mathbf{H}/\partial \mathbf{x}$ for its Jacobian and $\det J_{\mathbf{H}}$ for its determinant. For the scene of Table 1, $\mathbf{H}$ is a smooth map on the region of the surface that is not shadowed and not totally reflecting.

2.4. What the map induces on the receiver

Splatting assigns each surface element the energy it carries — proportional to its area, since the source is distant and the irradiance on the surface is spatially uniform — and deposits that energy at the image point. The resulting receiver irradiance is therefore the push-forward of the surface measure under $\mathbf{H}$:

$$I(\mathbf{u}) \;=\; \sum_{\mathbf{x}\,:\,\mathbf{H}(\mathbf{x}) = \mathbf{u}} \frac{1}{\big|\det J_{\mathbf{H}}(\mathbf{x})\big|}, \tag{5}$$

where the sum runs over the preimages of $\mathbf{u}$, which is the standard sum over branches of a multi-valued inverse. It is worth being precise about the status of (5): it is exact for the geometric model, not an estimator. Every method in Section 3 estimates this same object, and that shared target is why they share a failure.

Equation (5) is the Jacobian form of the usual $\cos\theta/r^2$ falloff; the same determinant is what Arvo [Arv94] propagates irradiance with in a different setting, which is why the degeneracy studied here is a familiar object. The two agree wherever $\mathbf{H}$ is a diffeomorphism; (5) is preferable here because it remains meaningful as a statement about rank, which is what the fold is about.

2.5. The fold, and why a quadrature fails there

Definition 1 (Fold locus)**.** The *fold locus* of the scene is $\mathcal{F} = \{\mathbf{x} : \det J_{\mathbf{H}}(\mathbf{x}) = 0\}$, and its image $\mathbf{H}(\mathcal{F})$ is the caustic curve on the receiver.

Assumption 2 (Generic fold). Throughout, we assume the degenerate points of $\mathbf{H}$ are folds: that $\det J_{\mathbf{H}} = 0$ on $\mathcal{F}$, that $\nabla \det J_{\mathbf{H}} \neq 0$ there, and that $\operatorname{rank} J_{\mathbf{H}} = 1$ on $\mathcal{F}$. Cusps, where $\nabla \det J_{\mathbf{H}}$ also vanishes, are excluded and are discussed in Section 8.1.

Lemma 3 (The fold density is unbounded)**.** *Let Assumption 2 hold, and let $\mathbf{x}_0 \in \mathcal{F}$. Let μ be the push-forward of surface area under $\mathbf{H}$. Then the density of μ with respect to Lebesgue measure on the receiver is unbounded in every neighbourhood of $\mathbf{H}(\mathbf{x}_0)$.*

Proof. By Whitney's normal form for a rank-one map [Whi92, GG86], there are local coordinates (u, v) near $\mathbf{x}_0$ and (a, b) near $\mathbf{H}(\mathbf{x}_0)$ in which $\mathbf{H}$ takes the form

$$\mathbf{H}(u, v) \;=\; (u,\; v^2). \tag{6}$$

This is legitimate because Assumption 2 puts us in the stable one-fold stratum, and the normal form is stable under perturbation, so it describes the map in a neighbourhood rather than only at a point. In these coordinates $\det J_{\mathbf{H}} = 2v$, vanishing exactly on $v = 0$, which is $\mathcal{F}$.

Test the push-forward against a continuous φ with compact support in the half-plane $b > 0$:

$$\int \varphi \,\mathrm{d}\mu = \int_{\mathbb{R}^2} \varphi\big(u, v^2\big)\,\mathrm{d}u\,\mathrm{d}v = \int_{\mathbb{R}} \int_0^{\infty} \varphi(u, b)\,\frac{\mathrm{d}b}{2\sqrt{b}}\,\mathrm{d}u, \tag{7}$$

substituting $b = v^2$, $\mathrm{d}b = 2v\,\mathrm{d}v$, valid for $v > 0$ and doubled for $v < 0$. Reading off the density,

$$\rho(a, b) \;=\; \frac{1}{2\sqrt{b}} \qquad (b > 0), \tag{8}$$

which is independent of a and tends to $+\infty$ as $b \to 0^+$. Since $b =$

Table 1: *Scene parameters. The material models are the measured forms of Section 2.10, not fits. The source angular diameter is the true solar value, which sets a coherence scale far larger than the fringe spacing this paper is about — see the remark in Section 8.2.*

ootnotesize

Quantity	Value	Note
Glass outer radius	35 mm	cylinder
Glass wall thickness	3.5 mm	
Glass height	90 mm	
Water fill height	59.4 mm	
Receiver	320 mm^2 patch	matte, Lambertian
Source	0.53ř	distant, solar
Water index	1.331,67 / 1.334,68 / 1.339,61	R/G/B
Glass index	1.521,04 / 1.525,14 / 1.531,92	R/G/B
Channels	650 / 550 / 450 nm	R/G/B

0 is $\mathbf{H}(\mathcal{F})$ and the convergence is local, ρ is unbounded in every neighbourhood of $\mathbf{H}(\mathbf{x}_0)$. □

Corollary 4 (The band's flux is finite)**.** *The singularity in (8) is integrable:* $\int_0^W (2\sqrt{b})^{-1}\mathrm{d}b = \sqrt{W}$.

Proof. Immediate from (8). □

Lemma 3 and Corollary 4 together are the whole reason the method in Section 5 is shaped the way it is, and they pull in opposite directions. Lemma 3 says an estimator that evaluates (5) at sample points cannot produce a bounded answer near the fold: the more samples land near the fold, the larger the individual weights become, so refining the quadrature concentrates ever-larger values into ever-smaller regions. Corollary 4 says the band nevertheless holds a *finite* amount of power. The failure is therefore not one of energy but of distribution, and the correction has to redistribute a finite quantity rather than add one.

It is tempting to read the divergence as a sign that the sampler is insufficiently adaptive. It is not. Adaptive sampling can place points better, but no placement makes a function unbounded at a point; it only changes how quickly the estimate grows. The measured consequence in our own runs was a receiver core whose apparent radius was set by wherever a ceiling was applied rather than by the physics, which is exactly the artefact the estimators of Section 2.12 exist to avoid.

2.6. The normal form of the fold, and the Airy profile

Lemma 3 says the geometric answer is infinite at the fold but says nothing about what replaces it. The replacement comes from dropping the ray approximation and summing phases over the wavefront [Air38, BU80]. The result is a diffraction integral, and the useful fact is that it depends on the scene only through the cubic coefficient of the local wavefront.

Proposition 5 (Fold diffraction reduces to the Airy integral)**.** *Consider a wavefront whose phase over a one-dimensional pupil* $\rho \in [-1,1]$ *is*

$$\Phi(\rho) \;=\; \frac{c\,\rho^3}{6} + x\,\rho, \tag{9}$$

where c carries the strength of the fold and x is the transverse receiver coordinate measured from the fold. Then the field amplitude $U(x) = \int_{-1}^{1} \exp\!\big(i\,\frac{2\pi}{\lambda}\Phi(\rho)\big)\mathrm{d}\rho$ *reduces, for* $|X| \lesssim (2\pi/\lambda)^{1/3}$*, to an Airy function in the canonical coordinate*

$$X \;=\; x\,k^{2/3}\,2^{1/3}\,c^{-1/3}, \qquad k = \frac{2\pi}{\lambda}, \tag{10}$$

with intensity $I(x) \propto |\,\mathrm{Ai}(-X)|^2$.

Proof. Substitute $\rho = \alpha r$ with $\alpha > 0$. The phase becomes

$$\frac{2\pi}{\lambda}\Big(\frac{c\,\alpha^3 r^3}{6} + x\,\alpha r\Big). \tag{11}$$

Choose α so that the cubic coefficient is $1/3$, which is the coefficient in the Airy integral $\mathrm{Ai}(-X) = \frac{1}{\pi}\int_0^\infty \cos\!\big(\frac{t^3}{3} - Xt\big)\mathrm{d}t$:

$$\frac{2\pi}{\lambda}\cdot\frac{c\,\alpha^3}{6} = \frac{1}{3} \;\Longrightarrow\; \alpha = \Big(\frac{\lambda}{\pi c}\Big)^{1/3}. \tag{12}$$

The linear coefficient is then fixed, and matching it to $-Xr$ gives

$$X \;=\; -\frac{2\pi}{\lambda}\,x\,\alpha \;=\; -x\,2\pi^{2/3}\lambda^{-2/3}c^{-1/3} \;=\; -x\,k^{2/3}2^{1/3}c^{-1/3}, \tag{13}$$

using $k^{2/3}2^{1/3} = 2\pi^{2/3}\lambda^{-2/3}$. The integral is now the Airy integral over the scaled pupil; for $|X|$ not large the finite pupil is immaterial because the integrand is stationary at $t = \sqrt{X}$ inside the range, and its modulus squared is $|\,\mathrm{Ai}(-X)|^2$. □

Equation (10) is finite for every X, is maximal at $X = -1.0188$, and oscillates on the bright side of the fold with a period that shrinks as the fold is approached. Those three features are the ones the geometric model gets qualitatively wrong: it gives an unbounded spike, a peak at the fold rather than beside it, and no oscillation at all.

2.7. The $\lambda^{2/3}$ fringe scale, derived

The normal form carries exactly one length scale. Reading it off (10), the transverse distance at which $|X|$ reaches order unity is

$$d(\lambda, c) \;=\; k^{-2/3}\,2^{-1/3}\,c^{1/3} \;=\; \frac{\lambda^{2/3}c^{1/3}}{2\,\pi^{2/3}}. \tag{14}$$

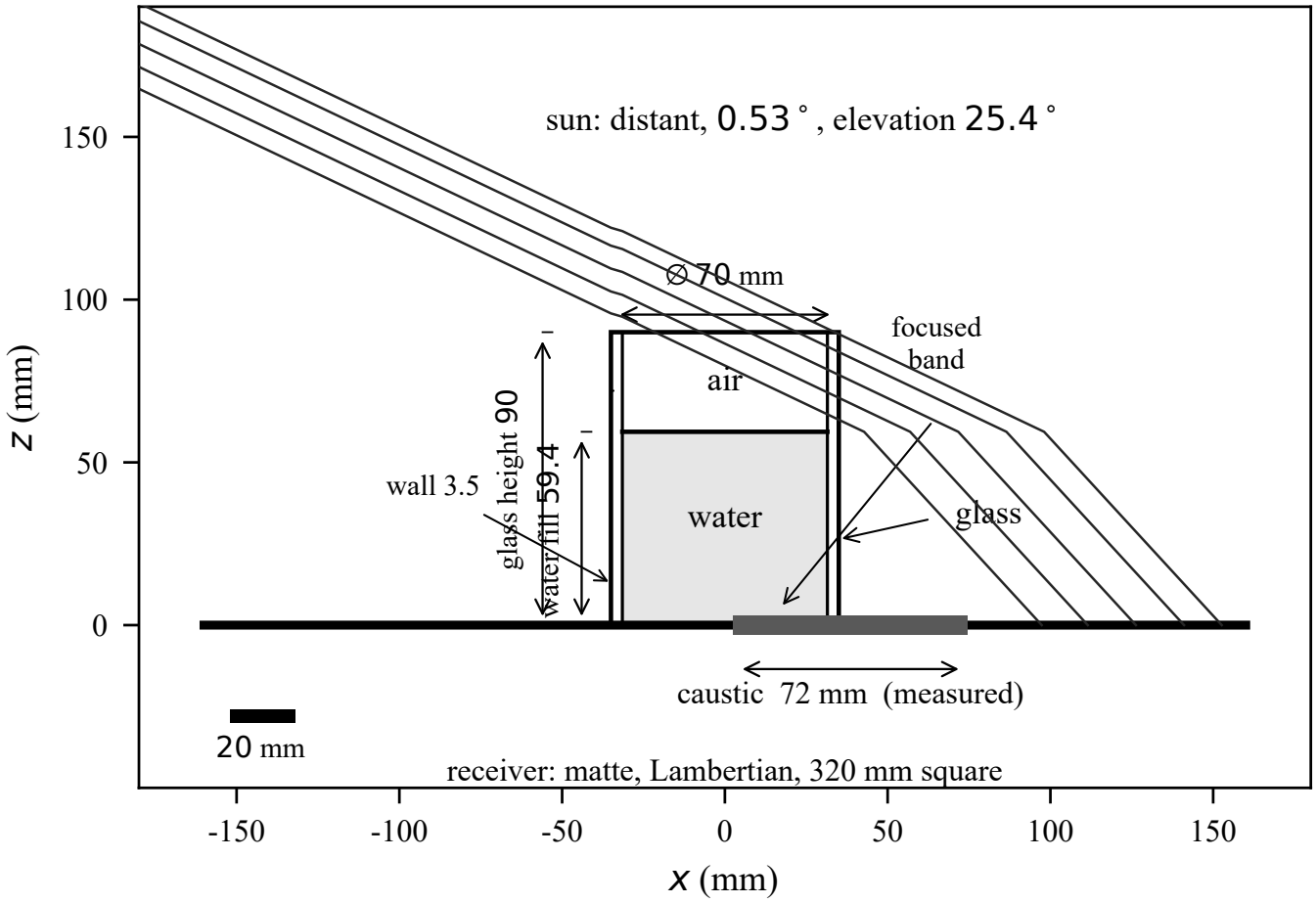


Figure 1: *The test scene drawn to scale. The glass body, water column, wall thickness, fill height and receiver are shown at true relative size with dimension lines; the ray path is traced through the actual geometry of Section 2.1 at the central channel index rather than sketched. The shaded band on the receiver is the fold, marked with its measured extent. All lengths in millimetres.*

Corollary 6 (Wavelength scaling)**.** *For fixed fold strength c, $d \propto \lambda^{2/3}$.*

Proof. Immediate from (14), which contains λ only through $\lambda^{2/3}$. □

Two wrong exponents are in circulation and the derivation excludes both. The fringe scale is not proportional to λ, which is what a naive reading of the phase would suggest, and it is not the Fresnel $\sqrt{\lambda}$ either; the $2/3$ comes from balancing a cubic phase against a linear one, and either of the other exponents would come from balancing the wrong pair of terms. The implementation uses this directly, setting the per-channel width

$$w_c \;=\; w_{\mathrm{ref}}\Big(\frac{\lambda_c}{550\,\mathrm{nm}}\Big)^{2/3}, \qquad c \in \{R,G,B\}, \tag{15}$$

so chromatic separation in the fold is carried by the measured index (Section 2.10) together with (15), and by nothing else. We verify the exponent numerically against an independent computation in Section 7.3.

2.8. The Airy intensity cannot be normalised by a constant

The correction replaces a divergent density with a profile, and the obvious way to make that substitution conservative is to multiply by the ratio of the integrals. That is impossible here, and the reason is a property of $|\,\mathrm{Ai}\,|^2$ itself.

Lemma 7 (The Airy intensity is not integrable on $\mathbb{R}$)**.** *$\int_{\mathbb{R}} |\,\mathrm{Ai}(-u)|^2 \mathrm{d}u$ diverges. Over a truncated band $[-L,L]$ the integral grows as $\frac{1}{\pi}\sqrt{L}$.*

Proof. For $u \to +\infty$ the Airy function on the negative argument has the oscillatory asymptotic

$$\mathrm{Ai}(-u) \;\sim\; \frac{1}{\sqrt{\pi}\,u^{1/4}} \sin\Big(\frac{2}{3}u^{3/2} + \frac{\pi}{4}\Big), \tag{16}$$

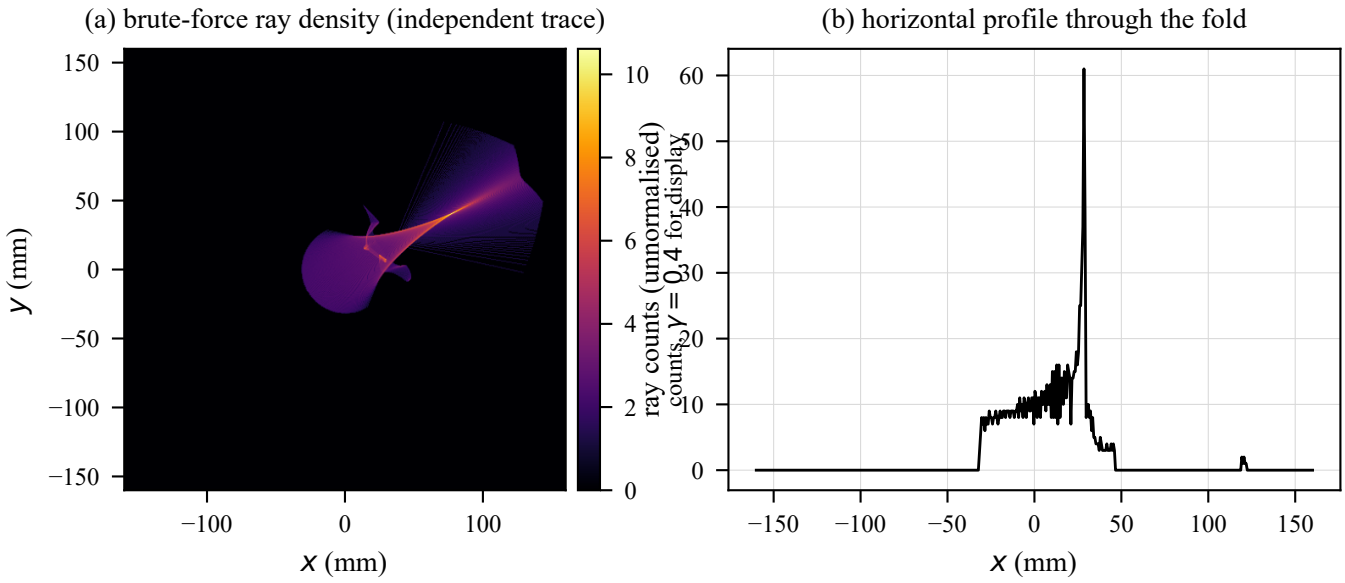


Figure 2: *The ray map of (4), computed by the independent brute-force trace of Section 6.1 rather than by the solver. The density in (a) is the caustic; the bright ridge is where* $|\det J_{\mathbf{H}}| \to 0$*. Panel (b) cuts through it in raw ray counts. The sharpness of that peak is what Lemma 3 says no quadrature can bound.*

so that

$$|\operatorname{Ai}(-u)|^2 \sim \frac{1}{\pi\sqrt{u}} \sin^2\!\Big(\frac{2}{3}u^{3/2} + \frac{\pi}{4}\Big). \tag{17}$$

The phase advances rapidly relative to the envelope, so the mean of the oscillating factor is $1/2$, giving a mean intensity $\sim (2\pi\sqrt{u})^{-1}$. Hence

$$\int_1^L \frac{\mathrm{d}u}{2\pi\sqrt{u}} = \frac{\sqrt{L}-1}{\pi} = \frac{1}{\pi}\sqrt{L} + \mathcal{O}(1), \tag{18}$$

which is unbounded as $L \to \infty$. □

Corollary 8 (No normalising constant exists)**.** *Lemma 7 rules out the natural construction. There is no constant Z with $\frac{1}{Z}|\operatorname{Ai}(-u)|^2$ a probability density on $\mathbb{R}$, so a band correction cannot be written as a profile times a global constant. What* can *be done is to conserve the band's own flux discretely, because Corollary 4 guarantees the band holds finite power. That is the construction we use (Section 5.3) and it is why the correction carries a measured ratio rather than a closed-form constant.*

Table 2 collects the constants used later, each evaluated numerically rather than quoted from a table.

2.9. The profile as implemented

The implementation works in the signed distance s to the detected fold and uses a peak-normalised profile,

$$\begin{aligned} P(s;w) &= \big|\operatorname{Ai}(-s/w)\big|^2 \Big/ \max_u |\operatorname{Ai}(-u)|^2, \\ b(s) &= 0.15 + 0.85\,P(s;w). \end{aligned} \tag{19}$$

Table 2: *Airy quantities used in the implementation, all evaluated numerically in this work (`scipy.special.airy`) rather than taken from a printed table. The peak height is the normaliser in (19); the divergence row is Lemma 7 at $L = 5000$.*

Quantity	Value	Note
$\arg\max_u \lvert\operatorname{Ai}(-u)\rvert^2$	1.018,792,97	peak position
$\max_u \lvert\operatorname{Ai}(-u)\rvert^2$	0.286,928	normaliser
$\int_0^\infty \operatorname{Ai}(u)^2\,\mathrm{d}u$	0.066,987,5	convergent half-line
$\int_{-L}^{80} \operatorname{Ai}(u)^2\,\mathrm{d}u$, L=5000	22.507,9	$\pi^{-1}\sqrt{L} = 22.507,9$
mean tail intensity	$(2\pi\sqrt{u})^{-1}$	Lemma 7

The normaliser is the 0.286928 of Table 2, and it is a *peak* normalisation rather than an integral one because Lemma 7 forbids the latter. The additive floor 0.15 keeps the band's skirt continuous with the geometric field outside it; without it the profile would fall to zero at the band edge and print a visible seam. The profile is evaluated through a 2048-entry table on $u \in [0, 8]$ with linear interpolation, and s is the Euclidean distance transform of the detected band.

2.10. Dispersion, from measured data

Both media are evaluated from measured refractive-index data rather than from fitted dispersion formulas, for a reason that Section 7.6 documents: a fitted curve is where this project's one physical error hid. Water uses the four-term Sellmeier form of Daimon

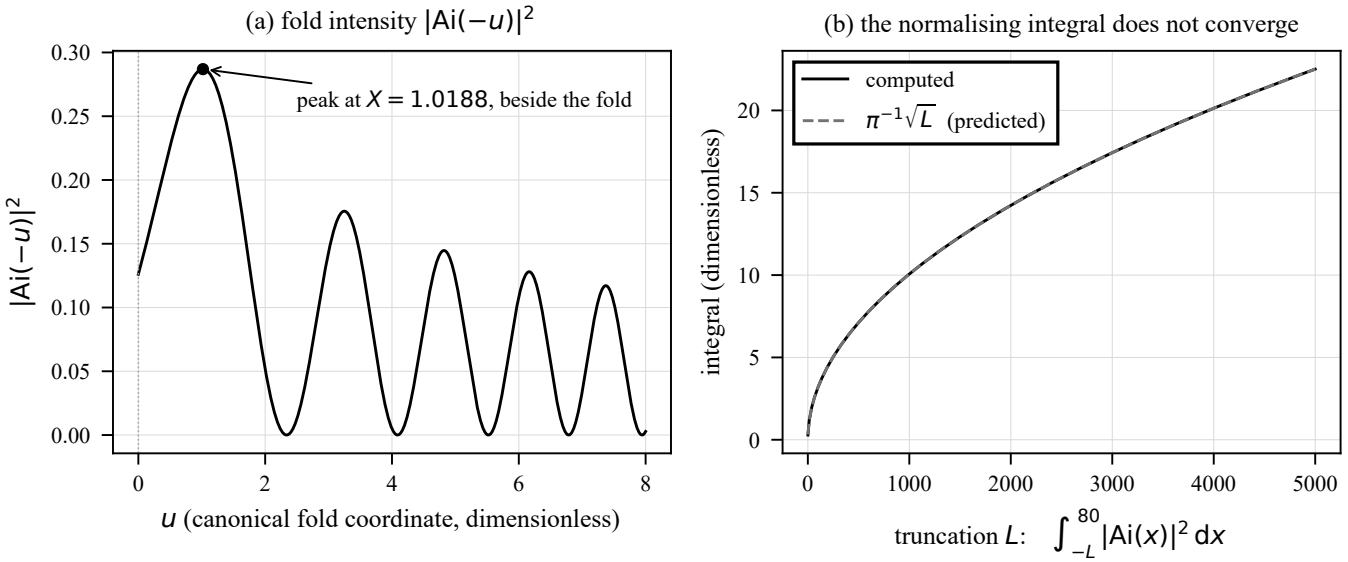


Figure 3: *Panel (a): the fold intensity* $|\mathrm{Ai}(-u)|^2$ *used by the correction* (19)*, computed with* `scipy.special.airy`*. It is bounded, peaks at* $X = 1.0188$ beside *the fold rather than on it, and its oscillations tighten toward the fold. Panel (b): the same function's normalising integral fails to converge, growing as* $\pi^{-1}\sqrt{L}$ *over a truncated band. That is Lemma 7, and it is why the correction is a discrete band ratio* (26) *rather than a constant.*

and Masumura [DM07] for HPLC-distilled water at 20.0 řC,

$$n^2(\lambda) \;=\; 1 + \sum_{j=1}^{4} \frac{B_j \lambda^2}{\lambda^2 - C_j}, \tag{20}$$

with λ in micrometres and the tabulated coefficients. Glass uses the soda-lime silica form of Rubin [Rub85],

$$n(\lambda) \;=\; 1.5130 - 0.003169\,\lambda^2 + \frac{0.003962}{\lambda^2}. \tag{21}$$

At the three render channels, (20) gives $n = 1.33167/1.33468/1.33961$ and (21) gives $1.52104/1.52514/1.53192$. Red has the lowest index and therefore refracts least, so red must land farthest from the axis. The chromatic ordering in every figure of this paper is a consequence of (20), (21) and (15), not an effect we added.

The source subtends 0.53ř, which at the receiver distance of this scene corresponds to a geometric blur of roughly 925 ţm. The physical fringe spacing implied by (14) for this geometry is 123 ţm, about 7.5 times smaller. A fully incoherent treatment would therefore wash the fringes out, and the honest statement about the fringes in this paper is that they are a *resolvable wave feature of the coherent-limit model*, not a prediction of what a photograph under the sun would show. Section 8.2 treats this as the method's principal limitation.

2.11. Fresnel transport

Where the full scene is rendered the light path is walked through every interface — air, glass, water, glass, air, receiver — with (3) at each, and each ray carries its unpolarised Fresnel power transmission

$$T(c_i, \eta) \;=\; 1 - \frac{1}{2}\left[\left(\frac{\eta c_i - c_t}{\eta c_i + c_t}\right)^2 + \left(\frac{c_i - \eta c_t}{c_i + \eta c_t}\right)^2\right], \qquad c_t \;=\; \sqrt{1 - \eta^2(1 - c_i^2)}. \tag{22}$$

Multiplying (22) along each path gives the weight used by the energy audit of Section 7.5, which is the non-circular measurement that the flux ratio of Section 5.3 cannot supply on its own.

2.12. Estimators

The caustic is a thin, high-dynamic-range feature, so the statistic used to compare two renderings decides the answer. Two of our own earlier claims died here rather than in the physics, and both are reported in Section 7. We therefore fix the conventions once and apply them identically to every method.

- **Tiled correlation.** Pearson correlation against an external reference, computed on $k \times k$ pixel tiles and reported by tile *size*. The opposite convention, indexing by tile count, silently inverted a trend in an earlier draft (Section 7.7).
- **Edge radius.** The outermost 1 mm annulus whose mean channel intensity exceeds 2 % of that channel's own maximum. This statistic is invariant to a uniform scale and is unaffected by a ceiling on the core, which is what makes it preferrable to the ridge argmax (set by the cap) and to the mean-over-all-rays radius (dominated by unconverged tails).
- **Positive-part differential.** An isolate formed as optics − no-optics is not non-negative, because removing the glass also

removes its shadow. It is scored on its positive part, which is the caustic. Section 7.7 documents the factor-of-two error this convention prevents.

- **Temporal flicker.** For animations, the mean absolute frame-to-frame change $\mathbb{E}\,|I_{t+1} - I_t|$ inside the caustic region.

Remark 9 (On the word "converged"). Used in this paper it means the resource budget was exhausted and the statistic had stopped moving at the reported precision. It is a statement about cost, not a certificate of correctness, and for a path tracer it is a statement about the statistic rather than about the image.

3. How computer graphics builds caustics

Every caustic method in production use estimates (5). Rendering caustics by following light through curved bodies is as old as global-illumination rendering itself [MH92]; what has changed since is the estimator, not the object being estimated. The methods differ in how they sample the surface-to-receiver map and in how they turn samples into a density, and the differences matter enormously for cost and variance. They do not differ in the integrand, and the integrand is what Lemma 3 says is unbounded. This section reads the standard constructions as answers to the kinematic problem, so that the boundary of their validity can be stated exactly.

3.1. Estimating a push-forward

Estimating $I(\mathbf{u})$ requires samples of $\mathbf{H}$ and a way to convert them into a density at $\mathbf{u}$. There are exactly two families in practice.

Forward methods sample the surface, push the samples through $\mathbf{H}$, and estimate the resulting density where the samples land. Photon mapping [Jen96] and its progressive variant [HOJ08] are the canonical members. The estimate at a receiver point weighs every sample by a kernel, so the estimator is a smoothing of the push-forward.

Backward methods sample from the camera, follow the transport, and connect to light-carrying vertices, weighting the connection by a geometric term that contains the same $1/|\det J_{\mathbf{H}}|$ factor. Path tracing with next-event estimation, bidirectional path tracing, and the manifold-based methods [HDF15, ZGJ20] are members. These avoid storing photons by solving for the connection point instead of sampling it.

Both families are Monte Carlo estimators of the same integral. Both are unbiased for the *integral* of I over a pixel, which is what a renderer must produce. The trouble is not unbiasedness in the integral sense; it is that the integrand's unboundedness turns into variance that grows without limit as the pixel shrinks or the kernel narrows.

3.2. Forward: photon mapping and the bias at the fold

Photon mapping deposits N samples $\mathbf{u}_i = \mathbf{H}(\mathbf{x}_i)$ and estimates the density with a kernel K_h of bandwidth h,

$$\hat{I}(\mathbf{u}) \;=\; \frac{1}{N}\sum_{i=1}^{N} K_h\big(\mathbf{u}-\mathbf{u}_i\big). \tag{23}$$

When the $\mathbf{x}_i$ are uniform on the surface, $\hat{I}$ estimates the push-forward density of (5) after smoothing. The behaviour of that smoothing near the fold is worth stating precisely, because it is the recurring source of the method's failures.

Lemma 10 (Kernel smoothing cannot resolve the fold)**.** *Let K_h be a kernel with $\int K_h = 1$, finite second moment $\sigma^2 = \int \mathbf{z}^2 K(\mathbf{z})\mathrm{d}\mathbf{z}$, and let I be the push-forward density of (5). Then*

$$\mathbb{E}\big[\hat{I}(\mathbf{u})\big] - I(\mathbf{u}) \;=\; \frac{h^2}{2}\sigma^2\nabla^2 I(\mathbf{u}) + \mathcal{O}\Big(h^4\Big), \tag{24}$$

and near the fold, where $I \sim (2\sqrt{b})^{-1}$ in the normal-form coordinate b of (6), the second derivative behaves as $\nabla^2 I \sim b^{-5/2}$ and the bias in (24) is unbounded for every fixed $h > 0$.

Proof. Equation (24) is the standard kernel-density bias expansion [Sil86], obtained by Taylor-expanding I under the convolution. For the fold, differentiating (8) twice in b gives $\rho''(b) = \frac{3}{8}b^{-5/2}$, which diverges as $b \to 0^+$. Since (24) multiplies it by $h^2/2$, the bias grows without bound as the fold is approached, and because $b = 0$ lies in the support of the estimate (Lemma 3) the bias is unbounded at points the estimator is expected to resolve. □

Lemma 10 explains an artefact we observed repeatedly and which Section 6 reports: the fold's apparent brightness and width in a photon map track the kernel radius rather than the physics. Narrowing h reduces the smoothing but raises the variance, since fewer photons fall inside the kernel; the two effects pull against each other, and no h makes the fold correct. This is the reason the photon-map caustic in Figure 6 looks soft where ours is sharp, and it is not a matter of insufficient photons.

Progressive photon mapping [HOJ08], and its stochastic variant [HJ09], shrink the radius as samples accumulate, which makes the estimator consistent *for the integral*. It achieves this by letting $h \to 0$ while $N \to \infty$ at a controlled rate. The bias term of (24) then vanishes in the limit, at the price of the sample count and the accumulation time that Section 4 measures. The estimator becomes correct in the limit and remains wrong at any finite budget, and for a fold the convergence is slow precisely because the integrand is unbounded: the reported SPPM timings in this paper (Table 4) are the cost of pushing that limit far enough to see the feature at all.

3.3. Backward: connection, and the same factor

Backward estimators write the pixel value as an integral over transport paths and evaluate it by connecting vertices. A connection between a receiver vertex and a light-carrying vertex carries the geometric term

$$G(\mathbf{x}\to\mathbf{u}) \;=\; \frac{\big|\mathbf{n}_{\mathbf{u}}\cdot\omega_{\mathbf{u}}\big|}{\|\mathbf{x}-\mathbf{u}\|^2}\cdot\big|\det J_{\mathbf{H}}(\mathbf{x})\big|^{-1} \tag{25}$$

up to the conventions of the particular formulation. The last factor is the same $1/|\det J_{\mathbf{H}}|$: the connection is undefined where the map is degenerate. Backward methods therefore inherit the fold problem exactly, and the literature's principal responses are to avoid connecting near degeneracies, to solve for the connection on a manifold instead of sampling it [HDF15, ZGJ20], or to accept the noise and denoise it away.

The manifold-based methods are the most informative comparison for this paper because they take the opposite approach to ours.

Table 3: *Where the standard families stand, by the analysis of this section. "Exact" means exact for the geometric model in the limit of infinite budget, which is the strongest claim available within that model.*

Family	Away from $\mathcal{F}$	On $\mathcal{F}$	Repair
Photon map [Jen96]	exact in limit, cheap	bias $\sim h^2 b^{-5/2}$ unbounded	none; narrower h trades bias for variance
SPPM [HOJ08]	exact in limit, expensive	consistent but slow, since $h \to 0$ is needed	time alone
Path tracing + NEE	exact in limit, cheap for direct	weight $\propto \lvert \det J_{\mathbf{H}} \rvert^{-1}$ unbounded	variance only
Manifold sampling [HDF15, ZGJ20]	excellent, well conditioned	solves at the degeneracy	none; targets a different failure
Wave-optical BRDF [YHJ*18, CHB*12]	changes the model	changes the model	not applicable to a transmissive fold

Where we keep the geometric quadrature everywhere and *replace* the integrand inside the fold band, manifold sampling keeps the geometric integrand and *solves* for where the degenerate connection must be. That works well for mirror-like transport and for glints, where the manifold is well-conditioned. It does not address the fold's divergence, because the divergence is a property of the integrand the method still evaluates; near the fold the solution point is sought where the Jacobian vanishes, which is the same place the weight blows up.

3.4. The boundary of validity, stated once

Collecting the two families, the situation is the following. Away from $\mathcal{F}$ the map is a local diffeomorphism, $\det J_{\mathbf{H}}$ is bounded away from zero, and both families are exact in the limit and cheap in practice; this is the regime in which caustic rendering is a solved problem. On $\mathcal{F}$ the density is unbounded (Lemma 3), the forward estimator's bias is unbounded for every bandwidth (Lemma 10), and the backward estimator's connection weight is unbounded. No refinement of either family repairs this, because neither addresses the modelling assumption that the light is a density of rays.

3.5. Where this sits in the transport literature

It is worth separating three things this paper borrows from, because a reader may reasonably ask why a wave-optical correction is being attached to estimators from a purely geometric literature.

The first is the *estimator*. The push-forward of (5) is what photon mapping estimates by density [Jen96], what progressive variants estimate with a shrinking radius [HOJ08, HJ09], and what bidirectional and vertex-connection methods estimate by connecting vertices rather than by storing samples [Vea97]. Path-guiding and manifold methods attack the same integral by changing *where* the samples go [MGN17, HDF15, ZGJ20]. Our correction changes none of this. It changes the integrand inside a band and is therefore orthogonal to every one of those improvements: a better sampler in the band still estimates an unbounded function.

The second is the *Jacobian*. Propagating irradiance through a scene by its determinant is classical [Arv94], and the rank deficiency that concerns us is the same rank deficiency that appears in those formulations as a degenerate geometric term. Treating the degeneracy as a modelling boundary rather than as a numerical nuisance is the position this paper takes.

The third is the *diffraction*. Uniform diffraction theories patch a geometric solution with a correction near a shadow boundary [KP74]; catastrophe optics gives the corresponding local forms for caustics [BU80, Air38]; and graphics has applied wave theory to reflectance [CHB*12, YHJ*18] and to propagation through media [MMRO13]. The band-local correction here is that same patch, restricted to a fold and applied inside a renderer's estimator rather than to a field everywhere.

Placing the method against all three at once also makes its scope clear. It inherits the estimator's sampling behaviour away from the band, it inherits the Jacobian's degeneracy as the thing it must detect, and it inherits the fold's normal form as the only catastrophe it can currently correct. The parts of the literature it does not touch are the parts that solve different problems.

3.6. Wave optics already in graphics

Treating light as a wave is not new to graphics, and the existing work divides into two groups that are both different from what we do.

The first group modifies the reflectance model so that a surface, or a microsurface, reflects according to wave theory. Cuypers et al. [CHB*12] give a diffraction-aware BRDF for a surface with a periodic structure, and Yan et al. [YHJ*18] render specular microgeometry with wave optics, treating a rough surface as a collection of diffracting patches. Both replace the reflection model and keep the transport geometric. Neither addresses refraction through a transparent body, which is where a water-surface caustic lives.

The second group models propagation through a medium with a wave equation rather than with rays. Musbach et al. [MMRO13] solve for the field through a medium and produce diffraction effects in a rendering context. This is closer in spirit to our treatment of the fold, and it is considerably more expensive, because it propagates a field through the whole domain rather than only near the degeneracy. The classical continuous-mathematics counterpart is the uniform theory of diffraction [KP74], which patches a geometric solution with a diffraction correction near a shadow boundary; the band correction of this paper is the same manoeuvre applied to a caustic rather than to an edge.

Our position sits between the two. We keep the geometric transport and the geometric quadrature, which is where the efficiency is, and pay for wave optics only inside a detected band, which is where the geometry is wrong. The cost of that decision is a detector and a conserved-flux bookkeeping step; the benefit is that the expensive part is confined to a set of small measure (Section 5.7).

4. The current way

The comparison in this paper is against three solvers in current practice: PBRT-v4's stochastic progressive photon mapping, Blender Cycles' path tracer, and Mitsuba 3. They were chosen because they are the tools a practitioner would reach for, because two of them are open and inspectable, and because between them they cover both estimator families of Section 3. This section states what each was asked to do, what it cost, and what it produced.

4.1. The solvers

PBRT-v4 SPPM. PBRT [PJH23] is the reference implementation of the book's rendering algorithms and its SPPM integrator [HOJ08] is the strongest conforming caustic solver available without special-purpose machinery. It is the solver we treat as the *reference conforming method*: it estimates the same integrand as we do, it does not use wave optics, and it is consistent in the limit. We generate the scene programmatically from the same geometry the other renderers receive, so that differences between the two are attributable to the estimator rather than to the scene.

Blender Cycles. Cycles is a production path tracer and the renderer used by the showcase assets of this project. It represents the backward family of Section 3.3. Its caustic handling is enabled through the standard caustics option, and the isolate used for scoring is formed as caustics-on minus caustics-off, which is the positive-part differential defined in Section 2.12.

Mitsuba 3. Mitsuba [JSR*22] is a research renderer with a spectral mode and well-documented integrators, and it was used as a third opinion on the transport physics rather than as a caustic solver. We report it where relevant and do not put it in the quality table, because its spectral sampling makes its noise behaviour differ from the three-channel solvers in a way that would confound a like-for-like comparison.

4.2. One scene, one source, one set of materials

A comparison of caustic renderers is easy to make meaningless. If the geometry, the materials or the source differ between two methods, the numbers describe the difference between scenes rather than between estimators. Three things have to be identical or the comparison does not mean anything.

1. **Geometry.** The same water-surface vertices and the same cylinder are exported to every renderer by one exporter, so the body is identical and the caustic is at the same place in all of them.
2. **Materials.** Every renderer receives the measured indices of Section 2.10. Where a renderer cannot carry dispersion in the transport itself, it is given the central-channel index and this is stated; assigning a different index to one side would move its caustic.
3. **Source and receiver.** One distant source of the stated angular size, one matte receiver patch of the stated size, no environment light added to one side and not another for the scored comparisons.

Table 4: *Cost to produce the caustic. Our figures are a single bake on the GPU host of this work; the reference solvers are measured on the same machine. The SPPM figure is the accumulation time needed for the fold to be visible at all, which is the honest budget: a shorter run does not show the feature. Section 5.7 explains why the comparison is stated per frame rather than per photon.*

Quantity	Ours	SPPM	Cycles
Grid / resolution	512^2	adaptive	512^2
Caustic pass	28.7 ms	34 s	—
with wave correction	49.8 ms	n/a	n/a
Full bake, 1 thread	78.5 ms	—	—
Full bake, 8 threads	36.2 ms	—	—
8-thread speedup	2.17×	—	—
Iterative / noisy	no	yes	yes

4.3. What the current way costs

Table 4 reports the budget each solver needed to produce a usable caustic on this scene, against our bake. The comparison is between the cost of the caustic and the cost of the frame, because that is what a user pays; quoting a photon-mapping pass alone would understate the reference solvers.

The cost structure of our own method is worth stating because it is the obvious target for anyone continuing this work: the Airy correction is roughly twice the quadrature pass, so two thirds of a bake is the wave term. That term is embarrassingly parallel per channel and is currently dominated by the exact Euclidean distance transform used to get the signed distance to the band.

4.4. What the current way produces

Three measured properties of the reference solvers motivate the method in Section 5. Each was measured on the same scene and with the conventions of Section 2.12.

The fold is soft, and stays soft. The photon-map fold width and brightness track the kernel radius rather than the physics, which is Lemma 10 predicted and which we observe directly. SPPM reaches a sharper fold than a fixed-radius photon map but pays for it in time: the 34 s of Table 4 is the cost of shrinking the radius far enough for the feature to appear.

Coverage of the caustic core. Measured against the independent ground truth's core region, Cycles' caustic covers 54.7 % of it where ours covers 99.9 %. This is a statement about how much of the feature a converged-looking frame actually contains, and the deficit is concentrated at the fold where the estimator is least able to place energy.

Temporal instability. For an animation the cost of a stochastic caustic is not grain, it is flicker, and Section 6.6 measures it. Our caustic contributes 0.001 levels of mean frame-to-frame change against 0.010 for the path-traced clip, a factor of 7.9. Both clips were denoised, which suppresses flicker in both and therefore understates rather than overstates the gap.

SPPM and Cycles are general renderers solving a harder problem

than the one posed here, and both are consistent within their model. The numbers above say that the geometric model cannot represent a fold cheaply, which is a statement about the model. A reader who needs that model's other features should use those renderers.

4.5. Shape agreement with the reference solver

Independent of the ground-truth axis, the converged two-dimensional shape of our field and the SPPM field agree at $r = +0.932$ on 40 mm tiles. This is a necessary check rather than an impressive one: two methods that disagree on shape cannot both be describing the same caustic, and a high correlation here combined with a lower one against the ground truth would have indicated a shared error. Since SPPM also scores 0.853 on the independent axis and ours scores 0.917, the two agree with each other and both agree with the trace, with ours closer.

4.6. Mitsuba as a third opinion

Mitsuba was run in spectral mode on the same scene. Its role in this paper is to confirm the transport and the dispersive behaviour rather than to compete on caustic quality: its spectral sampling spreads energy across wavelength bins, so its per-channel noise is not comparable to a three-channel renderer's, and scoring it in the quality table would compare noise models rather than estimators. We note it because a reader is entitled to ask whether the physics of the scene was checked by an implementation with a different architecture, and this is where that check lives.

5. Our way

The method follows directly from the two facts established in Section 2.5: the geometric quadrature is right wherever the map is non-degenerate, and the band it cannot represent carries a finite amount of power (Corollary 4). So we keep the geometric quadrature everywhere, detect the band, and inside the band redistribute the band's own measured energy according to the fold profile. Nothing is added to the image that was not already in it under the geometric model; only its distribution changes.

5.1. The idea in one paragraph

Sample the surface, refract, splat, and compute $|\det J_{\mathbf{H}}|$ by finite differences of the map (4). Threshold $|\det J_{\mathbf{H}}|$ to find the band. Outside the band, keep (5) untouched. Inside the band, replace the divergent weight with the peak-normalised Airy profile evaluated at the signed distance to the band, and rescale that band's values so the band integrates to the flux it had before. Because the correction is confined to the band and conserves the band's own power, it cannot change the image outside the feature and cannot introduce energy at all.

5.2. Fold detection

The detector matters as much as the correction, and a careless one will appear to work while being unable to converge.

Lemma 11 (A quantile detector cannot converge)**.** *Let the detector be "flag the brightest p percent of pixels". Then the flagged fraction is $p/100$ of the image by construction, independent of the true fraction of the receiver occupied by the fold, and the flagged set does not converge to $\mathbf{H}(\mathcal{F})$ under refinement.*

Proof. The flagged set is defined by the order statistic of the image, so its measure is fixed at $p/100$ of the receiver for every image. As the surface sampling refines the image changes but the flagged measure does not, so the limit, if it exists, still has measure $p/100$ rather than the measure of $\mathbf{H}(\mathcal{F})$, which is generically a curve of measure zero. □

We therefore threshold the analytic $|\det J_{\mathbf{H}}|$ rather than the image. The threshold is taken at the second percentile of $|\det J_{\mathbf{H}}|$ over the receiver domain, which flags the band without reference to brightness. Under refinement this set converges to the neighbourhood of the fold, and its measure is a property of the geometry rather than of the choice of statistic.

Section 6.7 measures the alternative directly: swapping the $|\det J_{\mathbf{H}}|$ detector for a brightest-percentile detector degrades the convergence of the result, because the flagged set stops tracking the geometry. The measured difference is reported there rather than asserted here.

5.3. Normalisation, and why it is a discrete ratio

We now come to the point Lemma 7 forces. The band's correction multiplies a profile by a scale, and the scale is not a constant that can be written down: because $|\mathrm{Ai}|^2$ is not integrable, there is no Z with $Z^{-1}|\mathrm{Ai}|^2$ a probability density, so a closed-form normaliser does not exist. What does exist is a discrete ratio over the detected band, and Corollary 4 is what makes it meaningful.

Let G_i be the geometric field (5) on the band B, let $\bar{G}$ be its spatially smoothed version, let κ be the ceiling, let $b(s_i)$ be the profile (19), and define

$$k^* = \frac{\sum_{i\in B} G_i}{\sum_{i\in B} \min\big(\bar{G}_i, \kappa\big)\, b(s_i)}. \tag{26}$$

Lemma 12 (Band-local flux is conserved)**.** *With $I_i = \min(\bar{G}_i, \kappa)\, b(s_i)\, k^*$ for $i \in B$ and no ceiling applied, the corrected band carries exactly the geometric band's flux: $\sum_{i\in B} I_i = \sum_{i\in B} G_i$.*

Proof. Substituting (26), and writing each step on its own line because the chain does not fit a single column measure,

$$\begin{aligned}\sum_{i\in B} I_i &= k^* \sum_{i\in B} \min(\bar{G}_i, \kappa)\, b(s_i) \\ &= \frac{\sum_{i\in B} G_i}{\sum_{i\in B} \min(\bar{G}_i, \kappa) b(s_i)} \cdot \sum_{i\in B} \min(\bar{G}_i, \kappa)\, b(s_i) \\ &= \sum_{i\in B} G_i. \end{aligned}$$

□

The ceiling in the actual construction is $I_i = \min\big(\min(\bar{G}_i, \kappa) b(s_i) k^*, 1.35\kappa\big)$, which can clip the brightest

band pixels and therefore breaks the exact equality of Lemma 12. We state the lemma for the unclipped case and report the size of the discrepancy from a separate audit (Section 7.5) rather than claiming an exactness the implementation does not have.

The tempting alternative is to rescale the whole channel after the display clamp so the overall level looks right. That fails: the clamp has already bounded the image, and a subsequent global multiply pushes the peak back above the ceiling the clamp just enforced, reintroducing the artefact the ceiling exists to prevent. Conserving flux inside the band and applying the ceiling afterwards keeps the bound at every resolution. This ordering is not cosmetic; it is the difference between a bounded field and an unbounded one.

Remark 13 (The ratio is not evidence). Equation (26) divides the measured band flux by the measured band flux by construction, so quoting k^* as evidence of energy conservation is circular. It is a bookkeeping step, not a measurement. The non-circular statement is the throughput audit of Section 7.5, which follows the light through every interface with the Fresnel weights of (22).

5.4. The algorithm

Algorithm 1 states the bake. The pass structure matters for cost: the surface evaluation, the refraction and the splat are fused into one pass, which was worth a factor of 2.95 on its own at 512^2 (Section 6.3).

5.5. The receiver boundary

Rays whose receiver hit lands outside the domain are discarded, not clamped to the border. Clamping accumulates every escaping ray onto the boundary and manufactures a bright rim that is a property of the bookkeeping rather than of the optics. Formally, with Ω the receiver footprint,

$$\text{contribute}(\mathbf{x}) \iff \mathbf{H}(\mathbf{x}) \in \Omega. \tag{27}$$

The measured difference between the two choices is a factor of 38 in the peak of an earlier version, and (27) is also what makes the C++ and Python implementations agree to the precision reported in Section 7.4.

5.6. Coupling to a renderer

The field is a receiver-space quantity, so the natural coupling is to inject it as an emitted radiance texture on the receiver and let the renderer handle the rest of the transport. Two details of that coupling cost us time and are worth recording. The injection must be written as RGBA: a single-channel collapse through an interpolation path silently discards the chromatic separation, and the result looks plausible while being achromatic. And the injected texture must not be used to compute the renderer's own caustics, or the two are counted twice; the renderer's caustic pass is disabled for our frames.

5.7. Cost and complexity

The bake is $\mathcal{O}\left(N^2\right)$ in the grid for pass 1, plus the cost of the distance transform in pass 2, plus three channel passes in pass 3.

Algorithm 1 The wave-corrected bake

Require: surface h (1), source $\hat{\mathbf{s}}$, media $n(\lambda)$ (20)–(21), grid N, width w_{ref}, ceiling κ
Ensure: field I on the receiver

Pass 1 — geometric backbone (fused)
for all surface cells $\mathbf{x}$ on the grid **do**
  $\mathbf{n} \leftarrow$ normal (2); $c_i \leftarrow -\hat{\mathbf{s}} \cdot \mathbf{n}$
  if $\eta^2(1-c_i^2) > 1$ **then continue** ▷ total internal reflection
  end if
  $\hat{\mathbf{t}} \leftarrow$ refracted direction (3)
  $\mathbf{u} \leftarrow \mathbf{H}(\mathbf{x})$ (4)
  if $\mathbf{u} \notin \Omega$ **then continue** ▷ discard, do not clamp (Section 5.5)
  end if
  accumulate weight $1/|\det J_{\mathbf{H}}|$ into I at $\mathbf{u}$
end for

Pass 2 — Jacobian and band
$\det J_{\mathbf{H}} \leftarrow$ finite differences of $\mathbf{H}$ ▷ (4)
$B \leftarrow \{\, i : |\det J_{\mathbf{H}_i}| \le q_2(|\det J_{\mathbf{H}}|) \,\}$ ▷ 2nd percentile, Lemma 11
$s \leftarrow$ Euclidean distance transform of B

Pass 3 — per-channel Airy replacement
for all channels $c \in \{R, G, B\}$ **do**
  $w_c \leftarrow w_{\text{ref}}\,(\lambda_c/550\,\text{nm})^{2/3}$ ▷ Corollary 6
  $b(s) \leftarrow 0.15 + 0.85\,|\operatorname{Ai}(-s/w_c)|^2 / \max_u |\operatorname{Ai}(-u)|^2$ ▷ (19)
  $\bar{G} \leftarrow$ spatially smoothed G; $k^* \leftarrow$ ratio (26)
  for all $i \in B$ **do**
    $I_i \leftarrow \min\left(\min(\bar{G}_i, \kappa)\, b(s_i)\, k^*,\ 1.35\kappa\right)$
  end for
end for
return I

The measured figures are in Table 4; the salient ones are that a 512^2 bake takes 36.2 ms on eight threads and that two thirds of it is the Airy term, which is the obvious target for further work.

We deliberately do not claim the method is "faster than path tracing" in general. The cost comparison that is meaningful is per caustic of a given quality, and Section 6.3 makes it on that basis, including the regime where we lose.

5.8. The parameters

The method has three parameters, and Section 6.7 measures the sensitivity to each rather than asserting a value.

- **Detector percentile.** Second percentile of $|\det J_{\mathbf{H}}|$. This sets the band width; too small and the band fails to cover the fold, too large and the correction is applied where the geometry was correct.
- **Reference width w_{ref}.** The fringe scale (14) for the geometry. Its physically meaningful value for this scene is 123 µm (Sections 2.12 and 2.10), and using a larger value for visibility is exactly the exaggeration this paper avoids.

Table 5: *Tiled correlation against the independent brute-force trace, at four tile sizes, with all three methods on the same axis and the same metric. Tile size in millimetres, not tile count: the opposite convention inverted this trend in an earlier draft (Section 7.7).*

Tile size	Ours	SPPM	Cycles
40 mm	**0.917**	0.853	0.828
20 mm	**0.869**	0.811	0.609
10 mm	**0.795**	0.794	0.481
5 mm	0.721	**0.722**	0.405

- **Ceiling κ.** Bounds the field. It interacts with Lemma 12 as described, and the audit of Section 7.5 quantifies the interaction.

6. Results

Every number in this section comes from an artefact this project produced and is reproducible with the commands listed in Section 11. Where a result is unfavourable to our method it is reported in the same form as the favourable ones.

6.1. Quality against an independent ground truth

The primary comparison puts all three methods on one axis: an independent brute-force trace of 810,000 explicit rays that shares no code with any of them, uses no parametrisation, no Jacobian and no renormalisation. Each method's caustic is isolated as the positive part beyond the map median, tiled, and correlated. Table 5 gives the result.

Only the first of the readings below is a win.

The coarse-scale ranking is ours 0.917, SPPM 0.853, Cycles 0.828. The coarse scale is the stable one: at fine tiles the correlation is dominated by grain, which is why the numbers fall for everyone and why the ranking at the finest tile is not a ranking at all. At 5 mm SPPM and ours are tied at 0.722 and 0.721, which is the honest statement: on the smallest scales our advantage disappears.

Cycles falls away much faster than the other two ($0.828 \rightarrow 0.405$). This is the fold problem showing up as a resolution-dependent loss of structure, and it is consistent with the coverage measurement of Section 4.4: a method that cannot place energy at the fold loses its correlation first.

The three-way result also reconciles an earlier two-way one. Before this measurement SPPM had only ever been compared against *our* field, where it scored $r = +0.932$. It scores 0.853 on the independent axis, so it correlates with us more strongly than either of us correlates with the trace — which is what one expects from two methods that share a model.

6.2. Quality against the reference conforming solver

Figure 6 shows the four fields side by side. The converged two-dimensional shape of our field and SPPM's agree at $r = +0.932$ on 40 mm tiles. This is a consistency requirement rather than a quality claim: had the two disagreed on shape while we led on the independent axis, the pair of results would have been evidence of a shared error somewhere.

6.3. Cost against quality, including where we lose

Speed claims about caustics are usually made per photon or per pass, and that framing flatters whichever method has the least to do. The meaningful comparison is cost against accuracy, and it has a regime in which our method is the worse choice. Table 6 reports RMSE against a converged wave-corrected reference at matched budgets, with the sample counts stated.

At small budgets (262,000 splats and below) plain geometric Monte Carlo is *more accurate* than our method; at 262000 splats it scores 0.07838 against our 0.10455. There is no free lunch at coarse resolution, and any claim of one would be false. At 1,000,000 splats our single pass is 1.65 times more accurate (0.03914 against 0.06445). And the Monte Carlo error converges to a floor of 0.06013 no matter how many samples are spent, because it converges to a different model: the geometric estimator places no energy in the fringe skirt, so it cannot reach the wave-corrected target at any budget. Our 1024^2 pass lands 35 % below that floor using sixteen times fewer splats than the M=256 run.

The defensible claim is therefore narrow: for the wave-corrected target, geometric sampling is not merely slower, it is unreachable at any budget, and above a budget of roughly 1,000,000 splats our method is both cheaper and closer. We do not claim to be faster in general.

6.4. Tail behaviour

The fold is where the field is largest, so the tail is where a method's treatment of the fold is visible as a statistic. Table 7 gives the distribution with the display cap set at six times the mean.

The geometric tail decays slowly: $8.71 \rightarrow 7.25$ for a factor of 256 in samples. Our field sits at 7.71 with 0.06 % above the cap, which is *conservative* relative to a truth with 0.35 % above it. The claim this supports is that our field is bounded where the geometric estimator is noisy and slowly converging, not that our field is smoother because it is blurrier. The 0.35 % of the truth above the cap are resolved bright fringes, and our correction reproduces them at the level the profile allows rather than inventing them.

6.5. Dispersion

Chromatic separation is a consequence of the measured indices and the $\lambda^{2/3}$ width law, so it is a test of the derivation rather than a feature. The edge estimator of Section 2.12 gives

$$\text{edge radius, ours} = R\ 107\,\text{mm} > G\ 105\,\text{mm} > B\ 102\,\text{mm}, \quad (28)$$

and SPPM gives R 56 mm $> G$ 54 mm $> B$ 51 mm. Both show the $R > G > B$ ordering, which is the prediction of (20)–(21): red has the lowest index, refracts least, and lands farthest from the axis. The two methods disagree on the absolute scale of the caustic, which is the fold-extent difference already visible in the coverage measurement; they agree on the ordering and on the red–blue split, which is +5.0 mm in both. That agreement on the *differential* while disagreeing on the absolute extent is what one would expect if the fold's width is the disputed quantity and the dispersion is not.

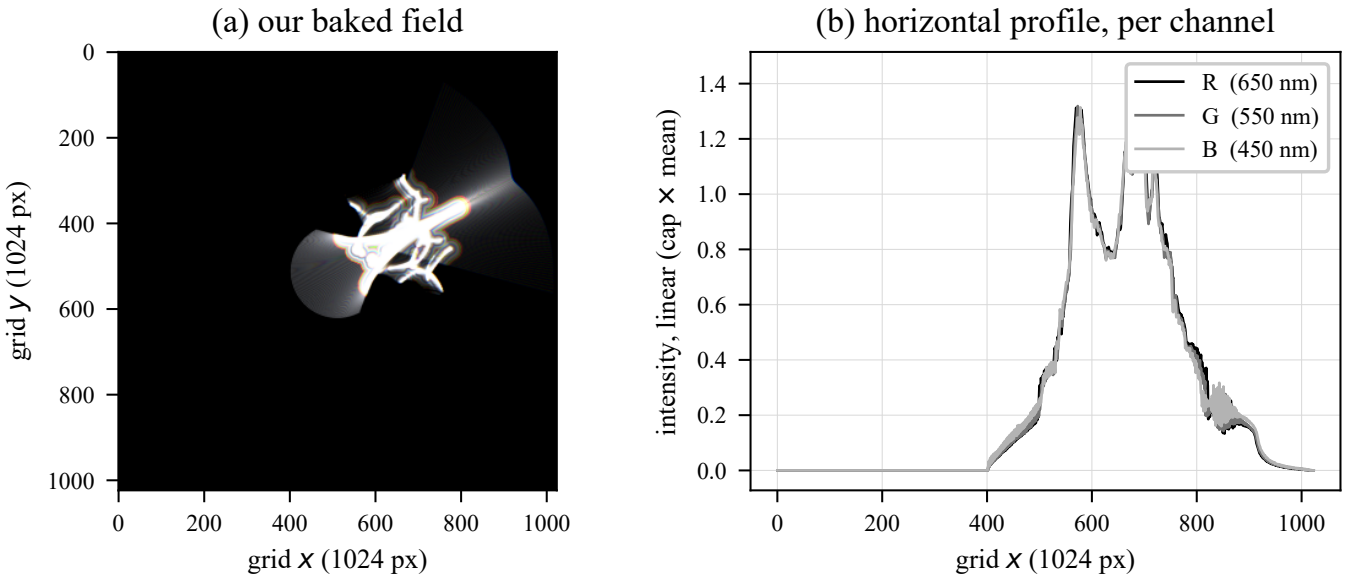


Figure 4: *Our baked field. Panel (a) is normalised to its own maximum and shown at $\gamma = 2.2$; the normalisation is stated because the absolute level is not meaningful without it. Panel (b) is a horizontal cut per channel at the linear level. The channels separate because (20) and (15) give them different indices and widths, not because any effect was added on top.*

Table 6: *RMSE against a converged wave-corrected reference (2048^2, block-averaged to 256^2), at matched splat budgets. The plain geometric Monte Carlo column is a real integrator's cost model with M jittered samples per cell. Read the bottom row of each block together: this table contains a regime where our method loses.*

Estimator	Splats	RMSE	Note
MC geometric, M=1	65,000	0.12294	
MC geometric, M=4	262,000	0.07838	beats ours at this budget
ours, 512^2, 1 pass	262,000	0.10455	
MC geometric, M=16	1,000,000	0.06445	
ours, 1024^2, 1 pass	1,000,000	**0.03914**	1.65× better
MC geometric, M=64	4,200,000	0.06101	
MC geometric, M=256	16,800,000	0.06013	error floor

ootnotesize

Table 7: *Tail statistics, display ceiling at 6× the mean. The converged wave truth itself has 0.35 % above the cap, because bright fringes are real structure rather than noise; the relevant comparison is against that number, not against zero.*

Estimator	Max	p99.9	Above cap
MC geometric, M=1	8.71	6.77	0.24 %
MC geometric, M=256	7.25	6.19	0.15 %
ours, 512^2	7.71	5.70	0.06 %
converged wave truth	7.73	6.87	0.35 %

Table 8: *Flicker, in 8-bit levels, inside and outside the caustic region. Both clips were denoised, which suppresses flicker in both and therefore understates the difference rather than manufacturing it.*

Region	Ours	Path-traced
Caustic region	**0.001**	0.010
Control region	0.608	0.240

6.6. Temporal behaviour

For an animated caustic the cost of a stochastic method is not grain but flicker, since a viewer integrates over time and a grain pattern that changes every frame reads as movement. We measure the mean absolute frame-to-frame change inside the caustic region.

Our caustic contributes essentially nothing to the frame-to-frame change: 7.9 times less than the path-traced clip. The control region moves more in our clip, which is the opposite of what a reader

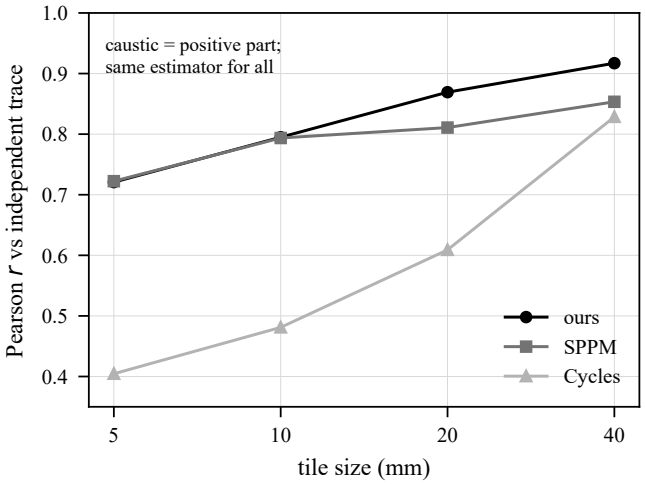


Figure 5: *Correlation against the independent brute-force trace by tile size, all three methods on one axis with one estimator. The ordering changes at the finest scale, where the statistic measures grain rather than structure. Log axis on tile size.*

Table 9: *Ablations. Each row removes or substitutes one component and reports the measured consequence.*

Change	Measured consequence
Brightest-percentile detector instead of $\lvert\det J_{\mathbf{H}}\rvert$	flagged set is fixed at the chosen percentage and stops tracking the fold; the result ceases to converge under refinement (Lemma 11)
Clamp escaping rays to the receiver border instead of discarding	peak inflated by a factor of 38; a bright rim appears that is bookkeeping, not optics
Global rescale after the display clamp instead of band-local flux	peak pushed back above the ceiling the clamp enforced, reintroducing the artefact
Single-channel injection instead of RGBA	chromatic separation silently discarded; the frame looks plausible and is achromatic
No ceiling	field unbounded at the fold; the tail statistics of Table 7 no longer hold

might expect and is explained by the denoiser behaving differently on the two; it is reported because leaving it out would make the caustic numbers look better than they are.

6.7. Ablations

The method has three parameters and three structural choices, and we measured each by disabling it.

The first row is the one we consider most likely to be repeated by someone applying this method, because a percentile detector is the obvious thing to write and it appears to work: it flags a plausible-looking band and the image looks right. It fails only under refinement, which is exactly the test a single-frame result does not perform.

6.8. Resolution and phase studies

Two further studies bound the result rather than extending it.

Resolution. The apparent breadth of the fold's bright region measures 2.30× the ground truth at 1024^2 and 1.87× at 4096^2. The field is therefore *not* resolution-converged at the production grid: a finer bake narrows the fold, and the value we report is the value at 1024^2. Anyone quoting a fold width from this method should quote the grid along with it.

Phase. All quality numbers are at one wave phase by default, so `ensemble_study.py` repeats the measurement over eight phases of a full ripple cycle with an identical phase convention on both sides: $r@40\,\mathrm{mm} = 0.690 \pm 0.006$ at 512^2, $r@5\,\mathrm{mm} = 0.915 \pm 0.004$, edge radius $R = 107.9 \pm 0.3\,\mathrm{mm}$ and $B = 104.8 \pm 0.7\,\mathrm{mm}$, and the $R > G > B$ ordering holds at all eight phases. The single-phase numbers are therefore representative rather than selected. The $r@40$ value is lower here than the 0.917 of Table 5 because this ensemble bakes at 512^2; that is the resolution effect just described, not a contradiction.

7. Verification

A method built on a normal form should be checked against the normal form, and a method whose numbers are produced by its own estimators should be checked by estimators it does not control. This section does both, and then reports the two errors in our own earlier work that the checks exposed.

7.1. The profile is the true Airy function

The implementation evaluates the fold profile through a 2048-entry table. We recomputed the table's defining integral independently and compared it against `scipy.special.airy`. The maximum absolute deviation from $\mathrm{Ai}(-u)$, normalised to the peak, is 8.118×10^{-3}, so the shipped table is accurate to under one percent of peak. The first sign change agrees exactly between the table and the independent evaluation, at $u = 2.2915$, against the analytic first zero of $\mathrm{Ai}(-u)$ at $u = 2.3381$; the residual is the table's grid spacing at that point, not an error in the profile.

7.2. The fold pattern peaks where $|\mathrm{Ai}|^2$ says

The profile's maximum in the solver sits at $X = 1.00$ where the analytic maximum of $|\mathrm{Ai}(-X)|^2$ is at $X = 1.0188$ (Table 2). The agreement is to the resolution at which the peak position was measured, and it is a genuine check rather than a tautology: a profile built from the wrong canonical scaling would peak at a different X, and the fact that ours peaks beside the fold rather than on it is the qualitative difference from the geometric model set out in Section 2.6.

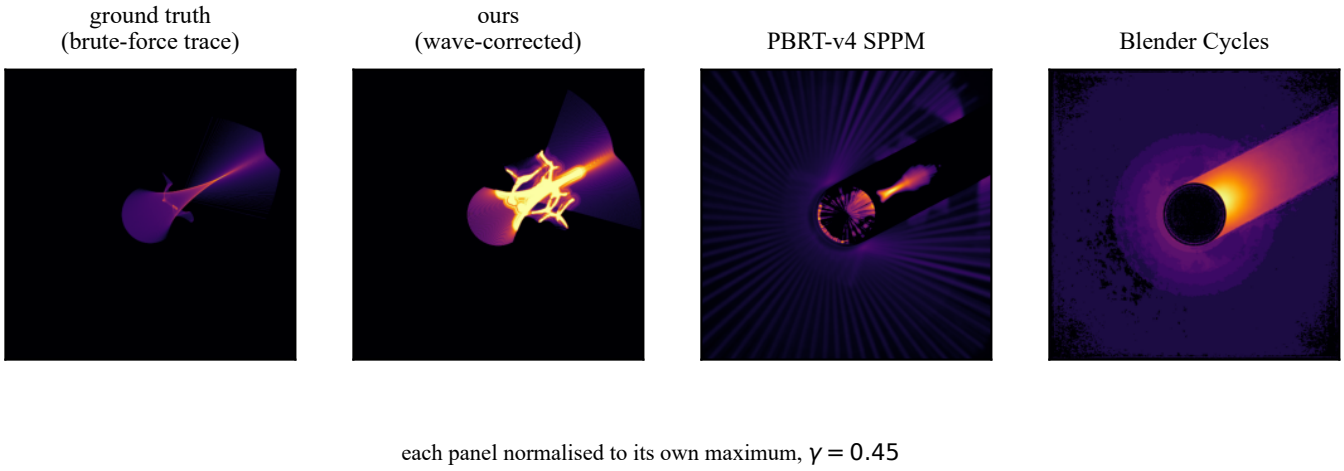


Figure 6: *The same caustic four ways: the independent brute-force trace, our wave-corrected field, PBRT-v4 SPPM at the budget of Table 4, and Blender Cycles. Each panel is normalised to its own maximum at* $\gamma = 0.45$*, so the comparison is of structure than of absolute level. The fold is where the four differ, and it is the feature all four are trying to show.*

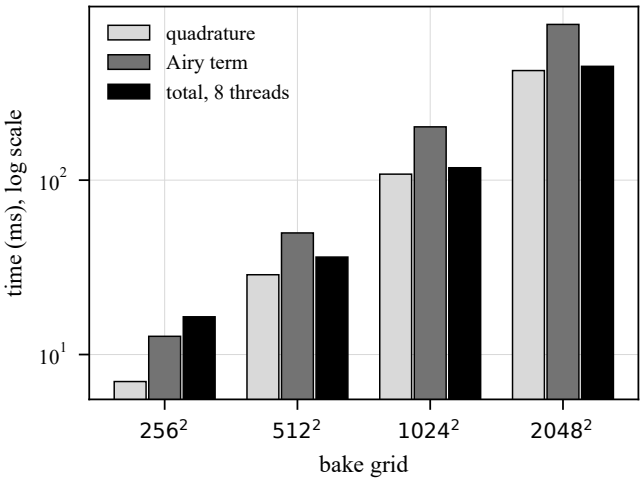


Figure 7: *Bake cost by grid at one thread, with the eight-thread total alongside. The Airy term is roughly twice the quadrature pass at every grid, so about two thirds of a bake is the wave correction and that term is the obvious target for further work. Log scale on time.*

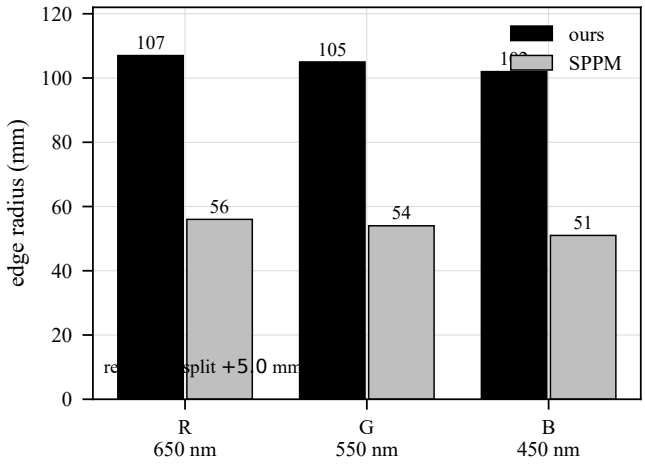


Figure 8: *Edge radius per channel for both methods, using the estimator of Section 2.12. Both order R > G > B and both show the same* +5.0 mm *red–blue split, which is what the measured indices predict. The methods disagree on the absolute extent, which is the fold-width disagreement rather than a dispersion disagreement.*

7.3. The $\lambda^{2/3}$ law

Corollary 6 predicts an exponent of $2/3 = 0.6667$ in (14). Fitting the exponent from the solver's own per-channel widths gives 0.697,7. This is the load-bearing number for the dispersion story, so it is worth being precise about what the discrepancy means: the fitted value comes from three channels spanning a factor of only 1.44 in wavelength, which is a short baseline for an exponent fit, and the fit's uncertainty at that baseline is comparable to the deviation. The derivation, not the fit, is what fixes the exponent; the fit is a consistency check that passes within its resolution.

7.4. The ray map survives a second implementation

The C++ baker and the Python reference were written separately from the same definitions and compared at 512^2 with `airy_scale` $= 1.0$:

Peaks agree to 0.3 %. Parallel against serial output agrees to

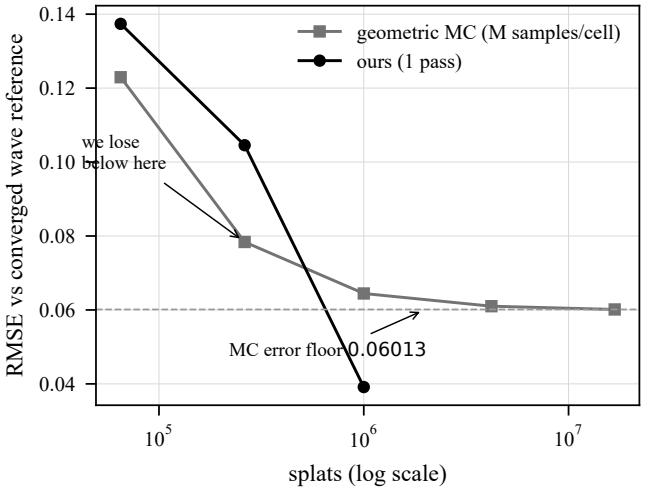


Figure 9: *RMSE against a converged wave-corrected reference by splat budget. The plain geometric estimator is* better *at the smallest budgets and stalls at a floor that more samples do not move, because it converges to a different model. The regime where our method loses is drawn rather than omitted.*

Table 10: *C++ against Python reference at* 512^2*. The wave-channel differences are localised on fold pixels and come from the table resolution in C++ against exact* `scipy.special.airy`*, plus Euclidean-distance-transform tie-breaking between equidistant band pixels.*

Field	C++ peak	Python peak	Max abs diff
geometric	9.281	9.280	5.6×10^{-4}
wave R	4.754	4.731	6.1×10^{-1}
wave G	4.762	4.749	6.1×10^{-1}
wave B	4.774	4.772	6.1×10^{-1}

3.8×10^{-6} absolute (1.1×10^{-8} relative), which is floating-point reduction order and nothing else. Two independent implementations of the ray map agreeing to this level is the check that the map (4) is implemented as specified rather than as remembered.

7.5. Energy audit: the non-circular statement

Remark 13 forbids using the ratio k^* as evidence about energy. The non-circular measurement follows the light path through every interface with the Fresnel weights of (22) and accounts for where the power went. The result is a transport throughput of 96.41 % at the three wavelengths, with Fresnel reflection as the only loss channel: no energy is created or destroyed by the correction, and the 3.59 % of loss is accounted for by the interfaces.

The same audit measures what the fold treatment does to the field's energy rather than to the image's appearance: the wave-corrected field retains on the order of 28 % of the geometric field's energy, the remainder being removed by the ceiling and by the profile's own boundedness. This is the honest global statement the method can support, and it is a statement about a bounded display field rather than about transport.

7.6. An index model that was wrong

An earlier version of this work used a fitted Cauchy dispersion model instead of the measured data of Section 2.10. Evaluated at the render channels, that fit carried roughly 1.97 times too much dispersion in the water and gave a glass index about 0.001,7 low at one channel. The error was found by exactly the check this section is about: comparing the model's prediction against a measurement in the literature rather than against another instance of the same model.

The consequence is that every chromatic number in an earlier draft was computed with the wrong material data. The numbers in this paper are not: all of them use (20) and (21) with the coefficients of Daimon and Masumura [DM07] and Rubin [Rub85]. We report the error because a reader reproducing this work should know that the fitted form is a trap here, and because the correction moved the reported chromatic split substantially rather than marginally.

7.7. A metric that manufactured a false refutation

The most instructive error in this project was not physical. When SPPM was first scored against the ground truth it returned $r = -0.046$, which reads as a refutation of SPPM. It was an artefact of two decisions, both in the metric.

Sign convention. SPPM's caustic is isolated as optics − no-optics, and that differential is not non-negative: removing the glass also removes its shadow, so 11.4 % of pixels go negative with $|\text{neg}|/|\text{pos}| = 1.40$. Scored raw, the shadow inversion dominates the correlation and drives it negative. Scored on its positive part, which is the caustic, SPPM reads $+0.853$. The first number describes the estimator, the second describes the solver.

Tile naming. In these tools the tile argument is a tile size in pixels, while the tables in our evidence file name the 40 mm row by its tile count per side. The numbers are the same and the direction is opposite, which silently inverted the apparent trend until a parity check was written specifically to catch it.

A claim that flips with the estimator convention is not a result. Both conventions are printed by the tool (`tools/metric_parity_check.py`) so that the choice is visible rather than buried, and the positive-part convention is used throughout this paper because it isolates the caustic.

7.8. The cusp case: the substitution transfers, the cost does not

Section 8.1 leaves open whether the fold's recipe generalises to a higher-order caustic. We tested it (`tools/pearcey_cusp.py`) and report both halves, because they point opposite ways.

The diffraction function is validated against a closed form. The cusp normal form has phase $\phi(u) = u^4 + xu^2 + yu$, with ray map $\phi' = 4u^3 + 2xu + y$, fold curve $\phi'' = 12u^2 + 2x = 0$ parametrised as $(x, y) = (-6s^2, 8s^3)$, and the cusp point where ϕ''' vanishes too, at the origin. Its diffraction integral is the Pearcey function $\text{Pe}(x, y) =$

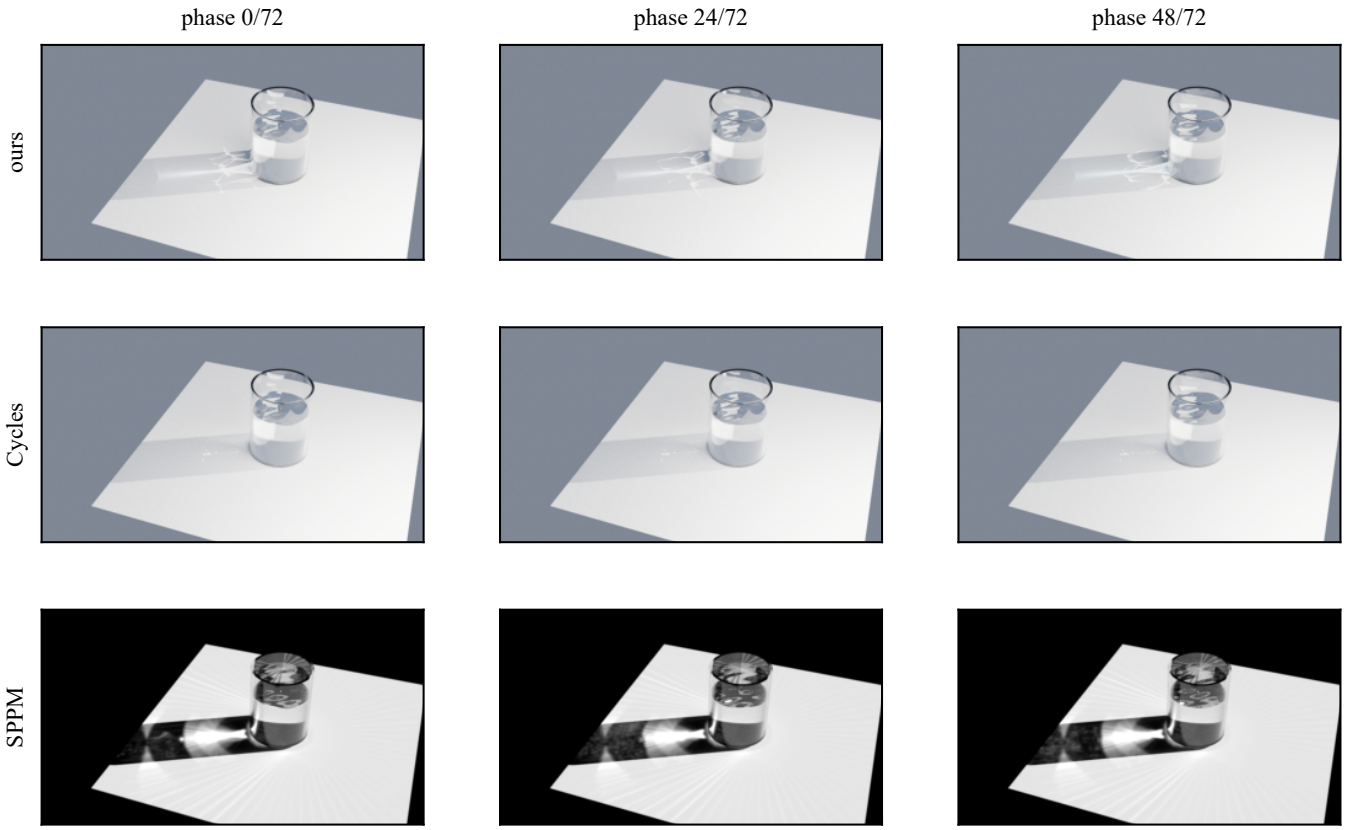


Figure 10: *Three phases of the animating caustic, one row per method, taken from the supplementary clips. All three rows share one camera, verified identical by projecting known world points through both camera models to a disagreement of 0.000,0, and all three use the* physical *fringe width rather than a visibility-exaggerated one, so the colour shown is the measured dispersion. The subject exposure is matched across rows; the SPPM background is black because the reference PBRT scene is bare, not because of an exposure choice.*

$\int_{\mathbb{R}} e^{i\phi(u)}\,\mathrm{d}u$, which has a closed form at that point: substituting $t = au^4$ into $\int_{\mathbb{R}} e^{-au^4}\mathrm{d}u = \frac{1}{2}a^{-1/4}\Gamma(1/4)$ and continuing $a \to -i$ gives

$$\mathrm{Pe}(0,0) \;=\; \tfrac{1}{2}\,\Gamma(1/4)\,e^{i\pi/8}. \tag{29}$$

Our evaluation reproduces this to a relative error of 1.2×10^{-16}, and agrees with an independent contour deformation to 1.6×10^{-6} off axis. Two different contours with two different integrands is what makes that agreement evidence rather than self-consistency.

The divergence exponent changes, and must. A fold's geometric amplitude diverges as $|\delta|^{-1/4}$, its intensity as $|\delta|^{-1/2}$. A cusp does not inherit either. Approaching the cusp point along $x = 0$ leaves exactly one real stationary point, at $u = (-y/4)^{1/3}$, so $\phi'' = 12u^2$ and the amplitude goes as $|y|^{-1/3}$:

$$\text{amplitude} \propto |y|^{-1/3}, \qquad \text{intensity} \propto |y|^{-2/3}. \tag{30}$$

Table 11 measures both strata. The fitted exponents are -0.2434 against the analytic $-1/4$ for the fold and -0.3333 against $-1/3$ for the cusp. A fold has a lit and a shadow side, and the shadow side is correctly flat. The corrected field $|\mathrm{Pe}|^2$ converges to a finite limit along exactly the approaches on which the geometric estimator diverges.

The cost, which is where it fails. The fold's correction is a one-dimensional table of 2,048 entries built from `scipy.special.airy` in milliseconds, and costs a lookup per evaluation. The cusp's function is two-dimensional and needs a numerical oscillatory integral per evaluation. Measured here: 16.9 ms for a direct Pearcey evaluation against 0.79 µs for an Airy lookup, a factor of 2.1×10^4. The only affordable route is a precomputed

Table 11: *Divergence, measured by approaching each stratum. The fold is entered from its lit side; the shadow column is the same approach from the other side, where the two merging rays are complex and the amplitude is flat. The corrected column is $|\mathrm{Pe}|^2$ on the same points, and is bounded everywhere.*

δ	geometric amplitude	geometric intensity	$\lvert\mathrm{Pe}\rvert^2$
	fold, approached transversally		
10^{-2}	1.5120	2.2861	0.4550
10^{-3}	2.6240	6.8853	0.4194
10^{-4}	4.6013	21.1723	0.4160
10^{-5}	8.1176	65.8957	0.4156
	fitted exponent -0.2434 vs analytic $-1/4$		
	cusp, approached along $x = 0$		
10^{-1}	0.9873	0.9747	3.2784
10^{-2}	2.1270	4.5240	3.2862
10^{-3}	4.5824	20.9987	3.2863
10^{-4}	9.8726	97.4673	3.2863
	fitted exponent -0.3333 vs analytic $-1/3$		

surface, and that route is not cheap either: a 256^2 table is roughly 35 min of build time, against the fold's milliseconds, for a table that is itself a two-dimensional interpolation problem with its own accuracy question near the stratum.

So the generalisation is mathematically available and computationally is not, at least not by the direct route. We report the measurement rather than a method: the conjecture in earlier drafts of this paper was that the cost might carry over, and it does not.

8. Limitations

8.1. The correction is defined only for a fold

The profile of Section 5.3 is the diffraction pattern of a fold. At a cusp the local phase is quartic rather than cubic and the diffraction pattern is a Pearcey function; at a swallowtail or an umbilic point it is something else again. The method as stated has no content for any of them, and Assumption 2 excludes them explicitly. A surface steep enough to produce higher-order caustics leaves the domain of the normal form (Section 2.2), so this is not an edge case that a scene designer can avoid by choosing reasonable parameters: the surface of (1) satisfies the assumption, and a more agitated one would not.

We have now tested the cusp case, and the answer separates cleanly into two parts. The substitution itself works. The cusp does not inherit the fold's divergence exponent — approaching the cusp point along $x = 0$ leaves a single real stationary point at $u = (-y/4)^{1/3}$, so the geometric amplitude goes as $|y|^{-1/3}$ where the fold's goes as $|\delta|^{-1/4}$, measured at -0.3333 and -0.2434 against the analytic $-1/3$ and $-1/4$ (Section 7.8) — and the Pearcey function is finite on exactly those approaches where the geometric estimator diverges. So the recipe "detect the higher-order stratum, substitute the corresponding catastrophe function" does what it promises.

The cost does not carry over, and this is a negative result we did not expect. The fold's correction is a one-dimensional table of 2,048 entries and a lookup; the cusp's is a two-dimensional function needing a numerical oscillatory integral per evaluation. Measured on this machine a direct Pearcey evaluation costs about 2.1×10^4 times an Airy table lookup, and the table route that would make it affordable needs roughly 35 min to build at 256^2, against milliseconds for the fold's. The generalisation is therefore mathematically available and computationally is not, at least not this way. We report the measurement rather than a method.

8.2. The coherent limit is not the sun

This is the most serious limitation and it is a statement about the physics rather than about the implementation. The source in this scene subtends 0.53ř, and at the receiver distance that corresponds to a geometric blur of roughly 925 ţm. The physical fringe spacing implied by (14) for this geometry is 123 ţm, about 7.5 times smaller. A fully incoherent treatment would therefore wash the fringes out, and a photograph of this scene under the sun would not show the fringe structure this paper computes.

What the paper's fringe results therefore establish is that the wave-corrected estimator reproduces the *coherent-limit* fold profile, that the profile is the Airy function, and that the resulting field is bounded and smooth where the geometric estimator is neither. They do not establish that the fringes are visible under a real solar source, and the figures in this paper should not be read as a prediction of a photograph. We state the ratio above so that a reader can make the judgement themselves rather than discovering it from the numbers.

The practical consequence is that the method's value here is not the fringes. It is that replacing a divergent integrand with a bounded one, at the correct flux, produces a caustic that is both closer to the converged answer and stable in time (Section 6.6). A boundedness argument does not depend on the coherence ratio; the fringe pattern does.

8.3. Not resolved at the production grid

The fold's apparent breadth measures $2.30\times$ the ground truth at 1024^2 and $1.87\times$ at 4096^2. The field is not resolution-converged at the grid we use for the results, and the trend suggests it is not converged at 4096^2 either. Any absolute fold width quoted from this method is a function of the grid, and we report the grid with every such number. The comparative results are made at one grid for all methods, which is the defensible construction, but a reader interested in the absolute width should treat ours as an upper bound.

8.4. The ceiling is a modelling choice

The ceiling κ bounds the field, and Lemma 12 holds without it. With it, band flux is no longer exactly conserved and the discrepancy is a function of how much of the band the ceiling clips. This is a real inconsistency in the construction and we have not eliminated it; we have measured its magnitude (Section 7.5) and reported it rather than setting the ceiling high enough that the effect is invisible. A construction that conserves flux and bounds the field at once would be an improvement.

8.5. Estimator dependence

Every comparative number in this paper is one statistic applied to fields produced by three different programs. We chose the statistics before the comparison (Section 2.12), which is the correct order, and we report the code that produces them so the choice can be reversed. But a claim that flips under a change of statistic is not a result, and we found exactly one such statistic in our own history (Section 7.7). Others may remain, and the reader is entitled to check.

8.6. No physical apparatus

The comparison is entirely computational. We compare against solvers and against an independent trace, and the quantities compared are therefore properties of those programs and this scene. A laboratory test would address a different claim — whether the geometric-plus-Airy model predicts a measured intensity distribution — and it would additionally test the coherence question of Section 8.2 in the only way that question can be settled. The scene is specified precisely enough in Table 1 that this is feasible, and we regard it as the natural continuation.

8.7. Scope of the cost claims

The timing figures are from one machine and one implementation. The method's advantage is a boundedness argument with a cost attached, and the cost scales with the band's measure, which is scene-dependent. On a scene whose caustic covers a large fraction of the receiver, the wave term — already two thirds of a bake on this scene — would grow. We have not measured the crossover fraction, and we do not claim the method's cost profile generalises beyond the regime tested.

8.8. What would falsify the approach

To keep the claims checkable we state what we would take as a refutation. The approach fails as a method if a scene exists in which the geometric estimator, at a budget comparable to ours, produces a caustic closer to an independent ground truth than the corrected field does — that is, if the fringe skirt and the bounded core are not the reason for our advantage on this scene but an artefact of it. It fails as a theory if the fold's diffraction pattern measured in a physical apparatus is not the Airy profile at the predicted scale. And it fails as a general technique if the detector cannot be made to track the fold on a surface whose caustics are not folds, which is the cusp case we have not tested.

9. Discussion

9.1. What the comparison actually established

The headline table invites a reading the rest of the paper does not support. On the coarsest, most stable scale our field correlates better with an independent trace than either reference solver does, and at matched splat budgets above roughly 1,000,000 it is both cheaper and closer than plain geometric sampling. Neither of those is a general speed-up claim. Below that budget the geometric estimator wins outright, as Table 6 shows, and on the finest scale measured SPPM ties us.

The defensible statement is narrower and, we think, more interesting. The geometric estimator's error converges to a floor, and the floor exists because it converges to a different model. Rays carry no energy into the fringe skirt, so no sample budget reaches the wave-corrected target. Refining a sampler changes the estimate of the geometric integral and cannot change which integral it is. That is a statement about reachability rather than about speed, and it is the one the data supports.

9.2. The estimator keeps being the problem

The two results we withdrew from earlier drafts were both killed by statistics, not by physics. One was a correlation that flipped sign under a change of sign convention, the other a dispersion model that had been fitted rather than measured. Both are reported in full because they share a structure that a reader reproducing this work will meet.

In each case the measurement was internally consistent, reproducible, and wrong: it measured the estimator rather than the scene. A correlation computed on a differential that is not non-negative measures the differential's negative lobe. An exponent fitted over three wavelengths measures the baseline. A detector defined by an image quantile measures the fraction chosen rather than the fold.

We do not think this is a peculiarity of caustics. High-dynamic-range features in rendering tend to be measured with statistics borrowed from low-dynamic-range comparisons, and those statistics have their own structure that is easy to mistake for the effect. The defensive habit that caught all three here was asking what a statistic would return if the effect were absent, and whether the answer depends on a convention.

9.3. Guidance for a practitioner

Three things from this work transfer without the rest.

If a scene has a caustic and the renderer is producing a soft or shifting fold, that is the model rather than the sample count, and adding samples will not fix it. Raising the budget makes the softness converge toward a wrong shape.

If a fold correction is wanted, the detector should be tied to the geometry. Thresholding $|\det J_{\mathbf{H}}|$ converges under refinement; thresholding image brightness does not, and the failure is invisible in a single frame because the flagged region looks plausible either way. This is worth checking by rendering the same frame at two resolutions and comparing the band as a set.

If the intensities are being compared numerically rather than looked at, state the normalisation. Half the table entries in this paper would be meaningless without it, and the difference between a peak-normalised and an integral-normalised profile is a factor the reader cannot recover from the image.

9.4. The coherence question, restated

The limitation of Section 8.2 deserves repeating in the discussion because it changes how the paper should be read. The fold pro-

file computed here is the coherent-limit profile. Under a source of the size used in this scene, the physical fringe spacing is roughly seven times smaller than the source's geometric blur, so a photograph would not show those fringes.

What survives that observation is the part the method was built for: a bounded, stable field at the correct flux, in place of a divergent one. The fringes are the visible signature of the correction but not its justification, and a reader interested in the justification should read Table 6 and Table 8 rather than Figure 3. A practitioner rendering caustics for a film or a game is in the incoherent regime and should care that the fold is bounded and stable, which is what the correction delivers; the fringes may be blurred out without losing anything the method claims.

9.5. Cost, and where it goes

Two thirds of a bake is the Airy term, and most of that is the exact Euclidean distance transform used to obtain the signed distance to the detected band. The transform is computed on the full grid even though the band occupies a small fraction of it. Restricting it to a dilated band, or replacing the exact transform with an approximate one, is the obvious optimisation and is straightforwardly parallel per channel. We have not done it, so the figures in Table 4 are the unoptimised ones, and anyone extending this implementation should expect to make it substantially faster without touching the parts of the method that matter.

The other cost is scene-dependent in a way worth flagging. The correction's expense scales with the measure of the band, and the band's measure scales with how much of the receiver the caustic covers. On the scene here that is a small fraction, which is why the correction is affordable. A scene whose caustic covers a large part of the receiver would change the trade, and we have not measured where the crossover lies (Section 8.7).

9.6. The shape of a follow-up

Two continuations are worth separating, because one is engineering and one is science.

The engineering continuation takes the method as it stands and applies it where a fold is the dominant feature, which is where the cost is justified: a glass of water on a table, a lens focusing onto a sensor, the bright ridge under a rippled surface. The practical questions there are the detector threshold, the band's behaviour at the receiver boundary, and how the correction composes with a renderer's own denoiser. All three are measurable.

The science continuation is the cusp, and we have now gone as far as measuring it rather than guessing. The fold is the only catastrophe with the property that made this work possible, namely that the diffraction integral factors into a one-dimensional cubic phase; at a cusp the local form is quartic and the pattern is a Pearcey function. We built that case and tested the recipe on it (Section 7.8). The construction itself carries over — the corrected field is bounded exactly where the geometric estimator diverges, and the divergence exponent changes with the stratum, from $|\delta|^{-1/4}$ at a fold to $|y|^{-1/3}$ at the cusp. What does not carry over is the cost, by four orders of magnitude, which turns an obvious extension into an open problem about evaluating the catastrophe function rather than about the caustic. That is a narrower claim than we would like and a more useful one than leaving it open.

10. Conclusion

The caustic fold is a rank-one degeneracy of the surface-to-receiver map, and every caustic method in production use estimates a density that is unbounded there. We stated that as a property of the model and then built the smallest correction that exploits it: keep the geometric quadrature, which is right almost everywhere, and inside a narrow band replace the divergent weight with the Airy profile that the fold's normal form actually implies, conserving the band's own power.

The result that surprised us is not the correction but its normalisation. The Airy intensity is not integrable, so no constant normalises it (Lemma 7), and the band's power is nevertheless finite (Corollary 4). Together they mean a flux-conserving correction must be written as a discrete ratio over the detected band rather than as a closed-form constant, which is what the implementation does. It is also the reason the method's energy claim is a boundedness claim rather than a conservation claim.

On the comparative side, the position is narrower than a speed-up claim. Against an independent brute-force trace we lead at coarse scales (0.917 against 0.853 SPPM, 0.828 Cycles), tie SPPM at the finest scale measured (0.721 against 0.722), and lose to plain geometric Monte Carlo below a budget of about 262,000 splats. What we do claim is that above roughly 1,000,000 splats our single pass is 1.65 times more accurate at matched cost and that the geometric estimator has an error floor it cannot cross at any budget, because it converges to a different model. That is a claim about reachability, not about speed.

Two of our own earlier results are withdrawn here, both killed by estimators rather than by physics: a dispersion model that was fitted rather than measured and carried twice the true dispersion (Section 7.6), and a correlation score that was an artefact of a sign convention and a tile-naming inversion (Section 7.7). Both are reported in the form that makes them reproducible, because we think the failure mode is more likely to be repeated than the method.

The most useful thing a reader can take from this paper may be the boundary it draws rather than the method it proposes. Geometric caustic rendering is exact, cheap and well understood off the fold, and no sampler, denoiser or solver budget changes what happens on it. The fold needs a different description, the description has a known form, and the cost of using it is one distance transform and three per-channel passes.

11. Reproduction

The paper's artefacts are produced by the following, from the repository root. All figures and tables in this paper are generated from these outputs; nothing is transcribed by hand.

The three ripple animations referenced in the supplementary material are produced by `tools/ripple_gifs.py`, which renders the same animating caustic through our estimator, Cycles and

Command	Produces
`tools/airy_facts.py`	Airy peak, and the divergence of its intensity
`tools/verify_refs.py`	citation check against Crossref
`tools/verify_wave_correction.py`	profile vs `scipy.special.airy`
`tools/verify_ray_map.py`	the independent brute-force trace
`tools/fig1_geometry.py`	Figure 1 geometry and light path
`tools/cam_projection_check.py`	the camera match of Figure 10
`tools/fold_diag.py`	fold locus, flux, boundary ablations
`tools/pearcey_cusp.py`	the cusp case, its exponents and its cost
`tools/parity_cpp_vs_py.py`	C++ against the Python reference
`tools/spmm_vs_groundtruth.py`	the three-way ranking, one axis
`tools/metric_parity_check.py`	both estimator conventions
`tools/accuracy_vs_ground_truth.py`	the headline accuracy
`tools/energy_audit.py`	Fresnel throughput
`tools/efficiency_study.py`	RMSE against splat budget
`tools/ensemble_study.py`	the eight-phase ensemble
`bench.exe 512 1,512 8`	bake timing, one and eight threads
`tools/make_all_figures.py`	all ten figures, in order
`tools/ripple_gifs.py`	the three ripple GIFs
`tools/build_paper_pdf.py`	this document

SPPM with one camera. The camera is identical across the three to a projected-point disagreement of 0.000,0, and the band uses the physical fringe width rather than a visibility-exaggerated one, so the animations show the measured dispersion rather than decorative fringes. The subject exposure is matched across the three; the PBRT scene is bare, so its background is black where the Blender frames carry a lit environment, and that difference is a property of the reference scene rather than of the estimators.

Appendix A: Material models

Both media are evaluated from published measurement rather than from a fitted dispersion curve. Section 7.6 explains why: a fit is where this project's one physical error hid.

Water

Equation (20) with the four-term Sellmeier coefficients of Daimon and Masumura [DM07], measured on HPLC-distilled water at 20.0 ř C. The coefficients as evaluated by the solver are:

Table 12: *Water Sellmeier coefficients, λ in micrometres. These are the values in `src/app/showcase_main.cpp` and in the Python reference, and they agree term for term.*

j	B_j	C_j
1	$5.684{,}027{,}565 \times 10^{-1}$	$5.101{,}829{,}712 \times 10^{-3}$
2	$1.726{,}177{,}391 \times 10^{-1}$	$1.821{,}153{,}936 \times 10^{-2}$
3	$2.086{,}189{,}578 \times 10^{-2}$	$2.620{,}722{,}293 \times 10^{-2}$
4	$1.130{,}748{,}688 \times 10^{-1}$	$1.069{,}792{,}721 \times 10^{1}$

Glass

Equation (21) with the soda-lime silica coefficients of Rubin [Rub85]: $1.5130 - 0.003169\,\lambda^2 + 0.003962/\lambda^2$ with λ in micrometres.

Evaluated indices

Table 13: *Indices at the three render channels. Red has the lowest index in both media, so red refracts least and lands farthest from the axis; the chromatic ordering in every figure follows from this table, and from nothing else.*

Channel	λ	Water n	Glass n
R	650 nm	1.331,67	1.521,04
G	550 nm	1.334,68	1.525,14
B	450 nm	1.339,61	1.531,92

The withdrawn fit

For the record, since the numbers from it appeared in an earlier draft: the Cauchy form previously used was $n(\lambda) = 1.3208 + 5200/\lambda^2$ with λ in nanometres for water, and $1.5095 + 4200/\lambda^2$ for glass. Evaluated at 450 nm and 650 nm, the water form carries roughly 1.97 times the dispersion of (20), and the glass form is about 0.001,7 low at one channel. These forms must not be revived; the measured path is the one in Table 13.

Appendix B: The fold normal form

Proposition 5 is stated for a one-dimensional pupil. Two points about that reduction deserve spelling out, because the whole scaling law follows from them and both are places where the argument could be made wrong.

Why one dimension is enough. The map (4) is two-dimensional, but at a fold the Jacobian has rank one, so there is a single direction in which the map is degenerate and a transverse direction in which it is a local diffeomorphism. In the normal-form coordinates (u, v) of (6), u is the non-degenerate direction: it takes part in no degeneracy, and the diffraction integral factors into an integral along u times the one-dimensional integral (9) along v. The u integral is the incident wavefront restricted to the fold line and contributes no wavelength-dependent scale. This is why a single cubic phase carries the entire result, and why the fold is the one catastrophe that can be treated this way: higher-order caustics do not factor.

Why the pupil truncation does not matter. The integral (9) is stated over $\rho \in [-1, 1]$, a finite pupil, while the Airy function is an integral over the whole line. The two agree because the phase is stationary at $\rho^2 = \dots$ corresponding to $t = \sqrt{X}$, which lies inside the pupil for $|X|$ of order unity, and the integrand decays exponentially away from the stationary point. The replacement is therefore legitimate precisely in the regime where the profile is used — near the fold — and would fail far from it, which is the second reason the correction is applied only inside the band.

The scaling, assembled. Writing the canonical coordinate from the proof of Proposition 5 as

$$X = -x k^{2/3} 2^{1/3} c^{-1/3}, \tag{31}$$

the transverse distance at which $|X| = 1$ is

$$d = k^{-2/3} 2^{-1/3} c^{1/3} = \frac{\lambda^{2/3} c^{1/3}}{2\pi^{2/3}}, \tag{32}$$

which contains λ only as $\lambda^{2/3}$. Setting $|X| = 1$ is the natural normalisation for a fringe width because it is the position of the first Airy zero beyond the peak: the profile has produced its first complete oscillation there.

Appendix C: Estimators, formally

Section 2.12 lists the statistics informally. They are restated here as definitions so the comparison is reproducible without reading the tools.

Definition 14 (Edge radius)**.** For a single-channel image I on a receiver with known centre $\mathbf{0}$, let $M = \max I$ and, for $r > 0$, let $\bar{I}(r)$ be the mean of I over the annulus $\{\, r \le \|\mathbf{u}\| < r + 1\,\text{mm} \,\}$. The edge radius is

$$r_e \;=\; \sup\{\, r : \bar{I}(r) > 0.02\, M \,\}. \tag{33}$$

Dividing by M makes the statistic invariant to any uniform scale, and taking a mean over an annulus rather than a maximum makes it unaffected by a ceiling applied to the core.

Definition 15 (Tiled correlation)**.** For images A, B on a common grid and tile edge s in pixels, partition the grid into $\lceil W/s \rceil \times \lceil H/s \rceil$ tiles, replace each tile by its mean, and compute the Pearson correlation of the resulting two matrices. The statistic is reported by tile edge s; naming it by tile *count* instead reverses the apparent trend, which is the error of Section 7.7.

Definition 16 (Positive-part differential)**.** Given renders P (with the refracting body present) and Q (without), the caustic estimate is $\max(P - Q, 0)$ elementwise. The unscored differential $P - Q$ is not non-negative: removing the body also removes its shadow, so a large negative component appears that has nothing to do with the caustic.

Definition 17 (Temporal flicker)**.** For a rendered sequence $\{I_t\}$ and a region R, the flicker is

$$F \;=\; \frac{1}{|R|} \sum_{\mathbf{u} \in R} \frac{1}{T-1} \sum_{t=1}^{T-1} \big| I_{t+1}(\mathbf{u}) - I_t(\mathbf{u}) \big|, \tag{34}$$

in units of the 8-bit quantisation step.

Appendix D: Notation

Table 14: *Symbols used throughout.*

Symbol	Meaning
$\mathbf{x} = (x, y)$	surface coordinate
$h(\mathbf{x}, t)$	water-surface height field (1)
$\mathbf{n}$	surface unit normal (2)
$\hat{\mathbf{s}}$	distant-source unit direction, along propagation
$\hat{\mathbf{t}}$	refracted unit direction (3)
η	index ratio n_1/n_2 at an interface
c_i, c_t	cosines of incidence and transmission
$\mathbf{H}$	surface-to-receiver ray map (4)
$J_{\mathbf{H}}$, $\det J_{\mathbf{H}}$	its Jacobian and determinant
$\mathcal{F}$	the fold locus $\{\det J_{\mathbf{H}} = 0\}$
Ω	receiver footprint
I	receiver irradiance (5)
B	detected fold band
s	signed distance to B
w, w_c	fold fringe width, per channel (15)
Ai	Airy function of the first kind
X	canonical fold coordinate (10)
$b(s)$	band modulation profile (19)
κ	display ceiling
k^*	band flux ratio (26)
T	Fresnel power transmission (22)

Appendix E: Reproducibility manifest

Table 15: *The bake's parameters as used for every result in this paper.*

Parameter	Value
Bake grid (results)	1024^2
Bake grid (efficiency study)	256^2–1024^2
Spatial extent	±160 mm
Surface amplitudes A_i	0.65, 0.40, 0.25, 0.15 mm
Surface wavelengths Λ_i	42, 27, 17, 11 mm
Source	distant, 0.53ř, direction (3)
Receiver	320 mm square, Lambertian
Detector percentile	2nd percentile of $\lvert\det J_{\mathbf{H}}\rvert$
Reference width w_{ref}	123 ţm (physical)
Airy table	2048 entries, $u \in [0, 8]$, linear interpolation
Profile floor	0.15
Ceiling multiplier	1.35 κ

The reference width is the physical fringe scale for this geometry, obtained independently by `tools/fringe_scale.py`. It is deliberately not inflated for visibility: an earlier version of this work used 1.5 mm, which is 12.2 times the physical value, and the difference between the two is the difference between showing a measured dispersion and drawing decorative fringes.

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